Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students

Project ID: R26-IT-150

Project Proposal Report

WIJERATHNA C D W – IT22123268

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Wijerathna C D W	IT22123268	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Sandani Dahanayake	18/03/2026	

ABSTRACT

Practical experimentation is fundamental to physics education, yet many schools in rural Sri Lanka lack adequate laboratory infrastructure, equipment, and technical support. Students are consequently limited to theoretical learning without sufficient practical exposure, restricting the development of scientific reasoning and conceptual understanding. Most existing digital learning platforms treat experiment simulations, quiz assessments, and performance monitoring as separate features, without integrating them into a unified adaptive learning environment.

This research proposes a mobile-based virtual physics laboratory application designed for Grade 6–11 students in rural learning environments. The system enables students to perform interactive physics experiments, complete lesson-based quizzes, and engage in gamified learning activities while receiving guidance from an AI-powered educational chatbot. A key contribution of the proposed system is the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module. This module integrates quiz results, virtual experiment performance, and lesson progression to analyze student learning, estimate student levels, detect weaknesses, and generate personalized feedback within the mobile virtual physics laboratory.

Each lesson is divided into sub-lessons assessed through both quizzes and virtual experiments. Quizzes are intelligently generated using MCQs and structured questions aligned with the Grade 6–11 curriculum, while experiments evaluate practical skills. The module displays separate scores for each sub-lesson and identifies weak areas. When low performance is detected, the system immediately assigns a personalized learning path containing explanations, guided examples, experiment procedures, and video materials to support deeper understanding of the weak sub-lesson. After engaging with the learning path, the student reattempts the quiz and experiments for that sub-lesson. If the student still performs poorly after reattempting, the system assigns additional external resources such as YouTube videos and reference materials for further knowledge building. Shuffled quiz retakes and structured progression prevent students from advancing without completing the current sub-lesson.

The system will be implemented using Flutter for mobile application development, Firebase Authentication for secure user management, Cloud Firestore for storing learning data, and a Python-based backend for analytics processing and adaptive feedback generation. The expected outcome is an accessible and intelligent mobile learning platform that enhances physics conceptual understanding while providing weakness detection, personalized feedback, and targeted remediation support for students in resource-limited educational environments.

Keywords: Virtual Physics Laboratory, Mobile Learning, Intelligent Learning Analytics, Adaptive Feedback, Weak Student Remediation, Personalized Instruction, Physics Education

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor for their continuous guidance, encouragement, and valuable feedback throughout the development of this research proposal. Their support and expertise helped me to clearly understand the research problem and improve the overall quality of this work.

I am also thankful to the academic staff of the Department of Information Technology at the Sri Lanka Institute of Information Technology for providing the knowledge, academic resources, and learning environment necessary to carry out this research study. Their teaching and support have greatly contributed to my academic development.

In addition, I would like to acknowledge my fellow project team members for their cooperation, discussions, and support during the early stages of this research work. Collaborative discussions helped in shaping the overall concept of the proposed system.

Finally, I would like to express my deepest appreciation to my family and friends for their constant encouragement, patience, and motivation throughout my academic journey. Their support has been invaluable in helping me successfully complete this research proposal.

TABLE OF CONTENT

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT 5

LIST OF TABLES 6

LIST OF FIGURES 6

LIST OF ABBREVIATIONS 6

LIST OF APPENDICES 7

1. INTRODUCTION 8

1.1 Background and Context 8

1.2 Literature Review 9

1.3 Research Gap 10

2. OBJECTIVE 11

2.1 Main Objective 11

2.2 Specific Objectives 11

3. METHODOLOGY 12

3.1 System Design Approach 12

3.2 Sub-Lesson Based Performance Monitoring 12

3.3 Detailed Performance Analysis and Weakness Detection 13

3.4 Adaptive Feedback Recommendation for Weak Students 13

3.5 Personalized Instruction and Feedback Generation 14

3.6 Learning Analytics Visualization 15

3.7 System Implementation Technologies 15

3.8 System Evaluation 16

4. HIGH-LEVEL SYSTEM ARCHITECTURE 17

5. USER REQUIREMENTS 20

COMMERCIALIZATION PLAN 22

7. BUDGET AND JUSTIFICATION 24

8. WORK BREAKDOWN STRUCTURE (WBS) 26

Task Breakdown per Member 26

9. GANTT CHART 31

10. REFERENCES LIST 32

11.APPENDICES 33

LIST OF TABLES

Table 1-development of tools cost 25
Table 2-Firebase services cost 25
Table 3-final budget 25
Table 4-WBS Table 27
Table 5-Risk analysis 35

LIST OF FIGURES

Figure 1-Methodology Diagram 16
Figure 2-Methodology Process 17
Figure 3-High-level system Archytecture 19
Figure 4-Gantt chart 31
Figure 5-UI sample mockups 35

LIST OF ABBREVIATIONS

Abbreviation	Description
MCQ	Multiple Choice Question
API	Application Programming Interface
FR	Functional Requirement
NFR	Non-Functional Requirement
UI	User Interface
NoSQL	Non-Relational Structured Query Language
SDK	Software Development Kit
JSON	JavaScript Object Notation
LMS	Learning Management System
STEM	Science, Technology, Engineering, and Mathematics

LIST OF APPENDICES

Appendix	Description	page
Appendix-A	Survey instruments	33
Appendix-B	Detailed Diagrams: Module architecture diagram	34
Appendix-C	Preliminary UI Mockups	34-35
Appendix-D	Risk Analysis	35
Appendix-E	Ethical Documents	36

1. INTRODUCTION

1.1 Background and Context

Physics education requires both theoretical understanding and practical experimentation to help students develop scientific reasoning and conceptual knowledge. However, many schools in developing regions lack adequate laboratory infrastructure, equipment, and trained instructors, which limits students' opportunities to perform experiments and observe scientific phenomena directly [1]. This challenge is particularly evident in rural educational environments where resources and laboratory facilities are limited.

To address these limitations, digital learning technologies such as virtual laboratories have emerged as promising solutions for improving access to practical science education. Virtual laboratory systems allow students to perform simulated experiments and interact with scientific concepts through digital environments without requiring physical laboratory equipment [2]. Research has shown that virtual experiments can enhance students' conceptual understanding and provide flexible learning opportunities that overcome time and location barriers in traditional education systems [1].

In addition to experiment simulations, modern educational platforms increasingly incorporate mobile learning technologies, quizzes, gamified activities, and intelligent tutoring systems to enhance student engagement and support interactive learning experiences. Mobile learning technologies allow students to access educational resources anytime and anywhere using smartphones or tablets, making them particularly useful for learners in remote or underserved areas [3]. Gamification techniques such as challenges, rewards, and progress indicators have also been shown to increase student motivation and participation in STEM education [5]. Furthermore, artificial intelligence technologies such as educational chatbots can provide automated explanations and learning guidance to support students during their learning process [10].

Despite these advancements, many existing educational systems treat experiment interaction, quiz evaluation, and learning analytics as separate components. Consequently, student performance data becomes fragmented and does not provide a comprehensive view of a student's conceptual understanding across lessons and activities [1]. To overcome this limitation, the proposed research aims to develop a mobile-based virtual physics laboratory platform that integrates experiments, quizzes, gamified learning activities, and AI-assisted learning support within a unified learning environment. Within this system, the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module plays a central role in monitoring student learning progress and generating personalized feedback based on integrated learning analytics.

1.2 Literature Review

Recent research highlights the increasing role of virtual laboratories and simulation technologies in science education. Virtual laboratory environments allow students to explore scientific phenomena through interactive simulations and digital experiments that support conceptual visualization and practical understanding of complex scientific concepts [3]. Studies have demonstrated that integrating virtual experiments into science education can significantly improve students' engagement and conceptual learning outcomes [11]. In particular, virtual experiments allow students to repeat experiments multiple times, explore scientific relationships interactively, and learn at their own pace in a safe digital environment [2].

Another important research area related to this study is learning analytics, which focuses on collecting and analyzing educational data to better understand student learning behavior and improve learning outcomes. Learning analytics enables educators and learning systems to monitor student engagement, identify learning patterns, and detect potential learning difficulties based on interaction data generated during educational activities [1]. Educational data mining techniques further support this process by analyzing large datasets generated from digital learning platforms to discover meaningful patterns in student learning behavior and academic performance [7]. Researchers have emphasized that learning analytics systems can help monitor student progress and support data-driven decision making in educational environments [6]. Research has also highlighted that personalised learning supported by big data and artificial intelligence presents significant opportunities for improving educational outcomes [9]. Recent work has also demonstrated that integrating AI-based performance prediction with learning analytics significantly improves student outcomes in online courses [15], and that equity-focused learning analytics in physics education requires structured guidance to be effective [17].

In addition to analytics-based evaluation, recent research has explored the role of artificial intelligence in providing personalized learning support. AI-based educational systems can analyze student responses and provide automated feedback to guide learners toward better conceptual understanding [10]. Studies have shown that leveraging AI in e-learning through personalized learning and adaptive assessment enhances student engagement and academic performance [16]. Recent studies have also demonstrated that combining interactive learning environments with AI-generated feedback mechanisms can significantly improve student learning outcomes and engagement in physics laboratory environments [14].

Mobile learning technologies also play a significant role in expanding access to education. Mobile learning platforms enable students to access educational content through smartphones and tablets, providing flexible learning opportunities that are particularly beneficial for students who cannot access traditional classroom environments [4]. However, many existing mobile learning applications focus mainly on content delivery or interactive activities without integrating structured performance evaluation mechanisms that combine theoretical assessments and practical experiment outcomes within a unified learning model.

1.3 Research Gap

Despite the increasing use of virtual laboratories and digital learning platforms in science education, several limitations remain in current educational systems. Many existing virtual laboratory platforms primarily focus on providing experiment simulations without evaluating how effectively students understand the underlying scientific concepts through those experiments. Although virtual experiments allow students to visualize scientific phenomena and interact with simulated environments, they often lack mechanisms to systematically monitor student learning performance across lessons and activities [3].

Another limitation in existing educational systems is that theoretical assessments and practical experiment activities are often implemented as independent learning features. Many digital learning platforms evaluate theoretical knowledge through quizzes, while experiment-based systems focus mainly on simulation interaction without linking experiment outcomes to student assessment data. As a result, these systems provide only partial insight into students' conceptual understanding and learning progress [8].

Furthermore, many existing learning analytics platforms are designed for large-scale learning management systems used in higher education environments. These systems typically require advanced technological infrastructure and may not be suitable for school-level learning environments or rural educational contexts where technological resources are limited [13]. Although virtual experiments and mobile learning platforms such as PhET simulations improve accessibility and engagement in science education [12], relatively few systems integrate experiment activities, quiz evaluation, learning analytics, and personalized feedback within a single mobile learning environment.

Recent research also highlights the importance of providing individualized feedback during experimental learning activities to improve conceptual understanding and learning performance in physics education [14]. Studies further demonstrate that integrating AI-based performance prediction with learning analytics improves student outcomes [15], and that personalized learning through adaptive assessment is essential for effective e-learning [16]. However, most existing virtual laboratory systems do not incorporate automated personalized feedback mechanisms based on integrated analysis of quiz results, experiment interactions, and lesson-based learning progress.

Therefore, a clear research gap exists in the development of an integrated mobile learning system that combines user management, quiz evaluation, virtual experiment performance analysis, and learning analytics to generate personalized learning feedback within a virtual physics laboratory environment. As noted by Grimm et al. [17], learning analytics in physics education still lacks structured guidance for equity-focused decision-making, reinforcing the need for adaptive systems. The proposed Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module addresses this gap by integrating quiz results, virtual experiment performance, and lesson progression to analyze student learning, estimate student levels, detect weaknesses, and generate personalized feedback in a mobile virtual physics laboratory.

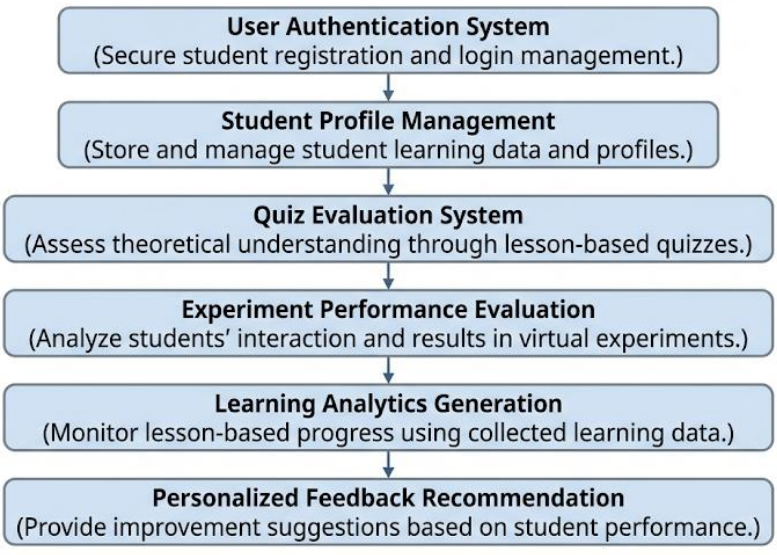
2. OBJECTIVE

2.1 Main Objective

To design and implement an **Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module** that detects weak students through quiz and experiment performance, assigns personalized learning paths for deeper learning, and provides additional resources for students who remain weak after reattempting assessments, within a mobile virtual physics laboratory application.

2.2 Specific Objectives

- Monitor student engagement in quizzes and virtual experiments at sub-lesson and topic levels.
- Generate intelligently designed quizzes covering Grade 6–11 curriculum content using MCQs and structured questions.
- Analyze student performance to detect weak areas in both theoretical and practical understanding.
- Generate personalized feedback messages and display improvement suggestions on the student home page based on detected weak areas.
- Assign personalized learning paths to weak students based on low quiz and experiment performance in specific sub-lesson areas, and enable reattempting of assessments after completing the learning path.
- Provide additional external resources such as YouTube videos and reference materials to students who continue to show low performance after completing the learning path and reattempting the quiz and experiment.
- Support repeated practice through shuffled quiz variations to encourage deeper conceptual understanding.
- Enforce structured learning progression by ensuring students complete each sub-lesson before advancing to the next..



3. METHODOLOGY

3.1 System Design Approach

The proposed system is designed as a **mobile learning platform** where students interact with physics learning materials through experiments, quizzes, and gamified learning activities. The Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module acts as the core analytical component of the system.

The system performs four main processes:

1. **Student Performance Monitoring**
2. **Weakness Detection and Analysis**
3. **Adaptive Feedback Generation**
4. **Personalized Remediation and Instructional Support**

These processes work together to monitor student learning behavior and generate personalized learning recommendations.

3.2 Sub-Lesson Based Performance Monitoring

The system is developed as part of the mobile virtual physics laboratory to manage student learning progress and evaluate performance in a more adaptive and supportive way. Each main physics lesson is divided into sub-lessons. For example, a lesson such as Force may contain sub-lessons like 1.1, 1.2, and 1.3. Each sub-lesson includes two assessment methods: quizzes and virtual experiments. The quizzes are intelligently generated for each lesson and sub-lesson, covering Grade 6–11 curriculum content, and include a combination of Multiple Choice Questions (MCQs) and structured questions to assess students' theoretical understanding at different cognitive levels.

For each sub-lesson, the app displays separate marks for quiz performance and experiment performance, such as Quiz = 75% and Experiment = 30%, and then uses these results to identify student strengths and weak areas. The quizzes measure students' theoretical knowledge, while the experiments evaluate practical understanding. Based on these two assessment results, the system analyzes performance at both sub-lesson level and main topic level.

- Correct and incorrect quiz questions per sub-lesson
- Virtual experiment performance and step completion status
- Identified weak theory and practical areas
- Personalized feedback messages and home page suggestions
- Experiment performance records
- Adaptive remediation resources and structured learning progression

All performance data is stored securely in Cloud Firestore and displayed through the student dashboard as feedback messages and learning suggestions on the home page.

3.3 Detailed Performance Analysis and Weakness Detection

A key feature of this module is its ability to provide detailed performance analysis rather than only showing marks. The system identifies which quiz questions were answered correctly and which had low scores, helping students understand their strong and weak theory areas. In the experiment section, the app evaluates practical performance and identifies which parts the student performed successfully and which parts need improvement. This analysis is summarized and presented through feedback messages and a suggestion section on the home page.

Theory Assessment – Quiz Analysis

Students complete intelligently generated quizzes for each sub-lesson, covering Grade 6–11 curriculum content. Each quiz includes MCQs and structured questions to assess theoretical understanding at different cognitive levels. The system evaluates answers, identifies incorrectly answered questions, and stores results with incorrect answer details and topic tags for further analysis.

Practical Assessment – Experiment Evaluation

Students perform virtual experiments within the mobile application. The system evaluates their practical performance and identifies which parts of the experiment were completed successfully and which parts need improvement. This analysis helps detect weak practical understanding at the sub-lesson level.

Lesson Progression and Structured Learning Enforcement

The module enforces structured learning progression by ensuring that students cannot move to the next sub-lesson without completing the current one. Lesson completion status is tracked and used along with quiz and experiment results to build a complete picture of the student's learning progress.

All collected data is stored securely in **Cloud Firestore**, allowing real-time synchronization between the mobile application and backend services.

3.4 Adaptive Feedback Recommendation for Weak Students

The module supports **adaptive feedback recommendation** for weak students. If a student receives a very low score in the experiment part of a sub-lesson, the system recognizes weak practical understanding and provides additional support such as simple explanations, guided examples, correct experiment procedures, and video-based learning materials.

If the student receives a low score in the quiz part, the system identifies incorrectly answered questions and provides **step-by-step explanations**, concept clarification, and supporting learning resources to improve theoretical understanding.

The module also supports repeated practice. When a student attempts a quiz again, the system presents a different shuffled set of MCQ and structured questions, ensuring variation and reducing memorization-based answering. This encourages deeper conceptual understanding of the lesson content.

Once the student engages with the assigned learning path and completes the provided resources, the system encourages the student to reattempt the quiz and experiment for that specific weak sub-lesson to demonstrate improved understanding. If the student still receives a low score after completing the learning path and reattempting the assessment, the system recognizes persistent weakness and assigns

additional external learning resources such as YouTube videos and reference materials specifically selected for that weak sub-lesson to provide further support for deeper knowledge building.

This approach allows the system to evaluate both theoretical knowledge and practical experiment performance together, generating targeted adaptive support for each student rather than treating assessments as separate learning activities.

3.5 Personalized Instruction and Feedback Generation

Based on quiz and experiment performance analysis, the system generates personalized instructional feedback and assigns structured learning support to students based on their identified weak areas. This transforms the mobile virtual physics laboratory into an intelligent learning support system that not only evaluates performance but also detects weaknesses and delivers targeted instructional support to enhance physics learning outcomes.

The personalized feedback and instructional support follow a rule-based decision model that evaluates quiz and experiment scores at sub-lesson level and responds in two distinct ways: immediate feedback messages and a personalized learning path assignment for weak students.

Immediate Feedback

After each quiz or experiment attempt, the system generates a feedback message specific to the student's performance in that sub-lesson. These feedback messages include:

- Sub-lesson quiz and experiment performance marks (e.g., Quiz = 45%, Experiment = 30%)
- Identification of which quiz questions were answered incorrectly, and which experiment steps were performed poorly
- A brief explanation of the weak concepts or practical areas detected
- A suggestion displayed on the home page encouraging the student to review the weak areas

The student cannot progress to the next sub-lesson until the current sub-lesson is satisfactorily completed, ensuring structured learning progression.

Personalized Learning Path Assignment for Weak Students

When a student receives a low score in the quiz or experiment of a sub-lesson, the system immediately recognizes the weak area and assigns a personalized learning path specifically designed for that student based on the weak lesson or sub-lesson identified. The system notifies the student and prompts them to accept the recommended learning path.

For example, if a student scores low in the Force lesson — Sub-lesson 1.1, the system detects this weakness and assigns a learning path specifically for Force 1.1. The student receives a notification suggesting the learning path and must accept it to proceed with the additional support.

Once the student accepts the learning path, it provides:

- Simple explanations of the weak theory concepts with worked examples relevant to the specific weak sub-lesson
- Step-by-step concept clarification for incorrectly answered quiz questions in that sub-lesson

- Correct experiment procedures with guided examples for poorly performed experiment steps
- Video-based learning materials aligned to the specific weak sub-lesson content
- Additional external learning resources for deeper understanding of the weak lesson topic

This learning path is personalized to each individual student based on their exact weak areas detected from both quiz and experiment performance. It is assigned immediately when weakness is detected, not after repeated attempts, ensuring that every student who struggles receives targeted instructional support at the right time. Once the student completes the learning path and engages with all provided resources, the system encourages the student to reattempt the quiz and experiment for that specific weak sub-lesson to demonstrate improved understanding. If the student still performs poorly after completing the learning path and reattempting the assessment, the system recognizes persistent weakness and assigns additional external resources such as YouTube videos and reference materials specifically selected for that weak sub-lesson, providing escalating instructional support until the student's understanding of the specific sub-lesson improves.

3.6 Learning Analytics Visualization

To help students understand their learning progress, the system displays performance results and feedback through the student dashboard in the mobile application. Flutter chart libraries and progress indicators are used to display information such as:

- Sub-lesson quiz and experiment marks (e.g. Quiz = 75%, Experiment = 30%)
- Identified strong and weak theory and practical areas per sub-lesson
- Personalized feedback messages and home page suggestions
- Lesson progression status and sub-lesson completion indicators

These visual indicators provide students with a clear overview of their quiz and experiment performance, weak areas, and learning progress across all physics sub-lessons and topics.

3.7 System Implementation Technologies

Several technologies are used to implement the proposed system:

- **Flutter** – Mobile application development framework used to create the student interface and interactive learning screens.
- **Firebase Authentication** – Used for secure student registration and login management.
- **Cloud Firestore** – Cloud-based NoSQL database used to store student profiles, quiz results, experiment records, and learning progress data.
- **Python Backend (FastAPI / Flask)** – Used to process learning analytics, evaluate performance data, and generate personalized feedback.
- **Flutter Chart Libraries** – Used to visualize learning analytics data through interactive graphs and progress indicators.

3.8 System Evaluation

The proposed system will be evaluated using both **functional testing and usability evaluation**.

Functional testing ensures that the system correctly performs tasks such as authentication, quiz evaluation, experiment performance tracking, and feedback generation.

Usability testing will assess how effectively students can interact with the application and understand the generated feedback and learning analytics dashboard.

Performance evaluation metrics include:

- accuracy of quiz score calculation
- correctness of experiment performance evaluation
- response time of analytics processing
- usability of the learning dashboard
- effectiveness of personalized feedback recommendations

These evaluation metrics will help determine whether the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module successfully supports structured monitoring of student learning progress within the mobile virtual physics laboratory environment.

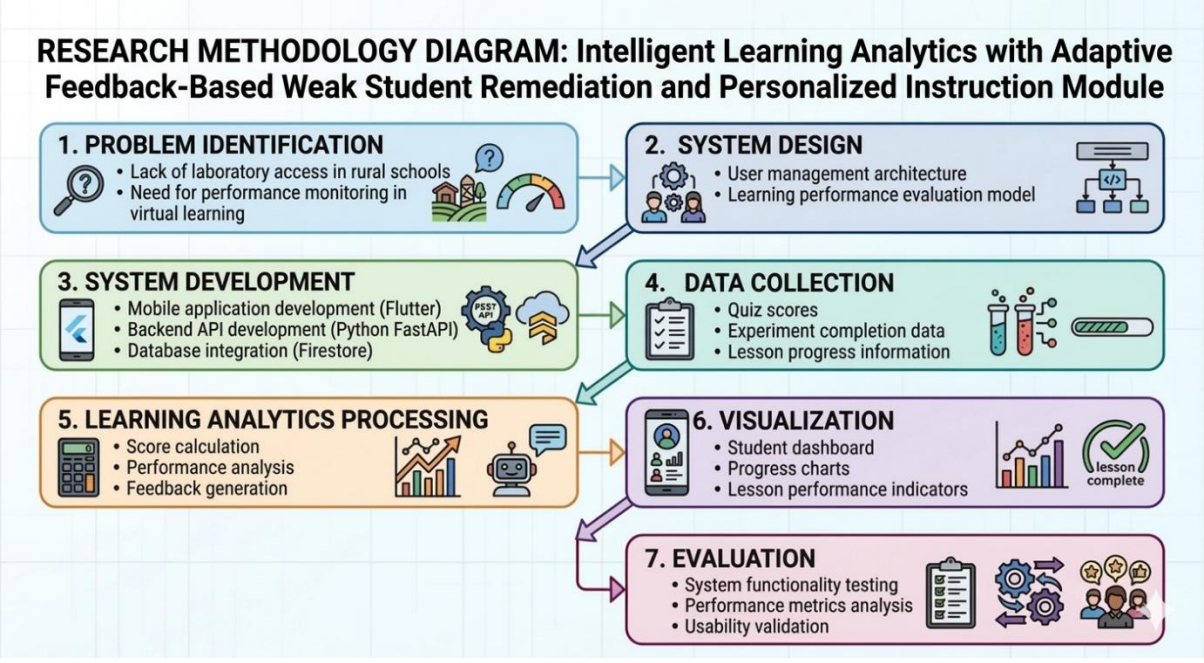


Figure 1-Methodology Diagram

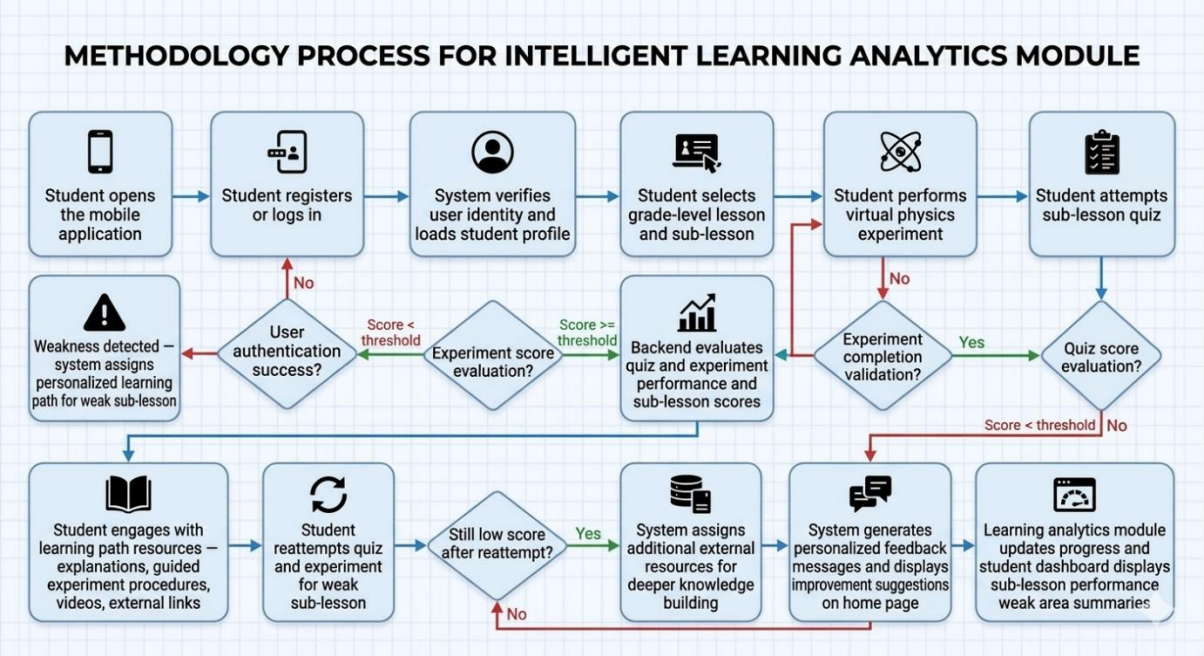


Figure 2-Methodology Process

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed solution is a mobile-based virtual physics laboratory application designed for Grade 6–11 students in rural Sri Lanka. The system combines four main functional components to support interactive and structured physics learning. These are the Grade-Adaptive AI Chatbot with Structured Learning Navigation, the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module, the Interactive Physics Experimentation and Learning Module, and the Intelligent Content Retrieval and Gamified Learning Generation Module. Together, these components create a unified mobile learning environment where students can access grade-based physics lessons, perform practical activities, answer quizzes, receive AI assistance, and monitor their lesson-wise learning progress.

At a high level, the system is organized into three layers. The first is the presentation layer, which is the Flutter mobile application interface used by students. This layer includes screens for login and registration, lesson selection, quizzes, virtual experiments, educational games, chatbot interaction, the learning progress dashboard, the personalized learning path interface for weak students, and the reattempt screen for students who have completed their assigned learning path. The second is the application logic layer, which contains the backend services responsible for quiz evaluation, experiment performance analysis, weakness detection, adaptive feedback generation, personalized learning path assignment, learning path completion tracking, reattempt evaluation, additional external resource assignment for persistently weak students, content retrieval, chatbot processing, and learning analytics generation. The third is the data layer, which stores student profiles, lesson data, quiz scores, experiment results, detected weak areas, assigned learning paths, learning path completion status, reattempt results, additional resource assignments, game records, and progress information using Cloud Firestore and related backend data services.

The Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module, which is the individual component of this research, plays a central

role in the system. It contains five main parts: user authentication, profile management, performance tracking, learning analytics generation, and personalized learning path assignment for weak students. The user authentication part manages registration, login, Google sign-in, and guest access through Firebase Authentication. The profile management part stores learner details such as name, grade level, and registration information. The performance tracking part records quiz results and virtual experiment outcomes for each sub-lesson, identifying which quiz questions were answered incorrectly and which experiment steps were performed poorly. The learning analytics part processes these records and generates sub-lesson level performance indicators, quiz and experiment scores, weakness summaries, and personalized feedback messages displayed on the student dashboard and home page suggestion section. The personalized learning path assignment part detects weak students based on low quiz or experiment scores in a specific sub-lesson and immediately assigns a targeted learning path containing simple explanations, guided examples, correct experiment procedures, video-based learning materials, and external resources to support deeper understanding of the weak sub-lesson content. Once the student completes the assigned learning path and engages with the provided resources, the system encourages the student to reattempt the quiz and experiment for that specific sub-lesson to demonstrate improved understanding. If the student still receives low marks after completing the learning path and reattempting the assessment, the system recognizes persistent weakness and assigns additional external learning resources specifically selected for that weak lesson topic to provide further support for deeper knowledge building. This escalating support mechanism ensures that every weak student receives continuous instructional support until their understanding of the specific sub-lesson improves.

The system components interact closely with one another. When a student logs into the application, the User Authentication part validates the identity and loads the student profile. The Learning Navigation and AI Chatbot component uses the student's grade level to guide the learner to appropriate lessons and provide level-appropriate explanations. The Interactive Physics Experimentation Module allows the student to perform virtual practicals and sends experiment completion and performance data to the backend. The Intelligent Content Retrieval and Gamified Learning Module provides supporting lesson content and lesson-based educational games. The Intelligent Learning Analytics Module receives quiz results and practical experiment data from these components, evaluates performance at sub-lesson level, detects weak areas, generates feedback messages, assigns personalized learning paths to weak students, tracks learning path completion, evaluates reattempt results, assigns additional external resources for persistently weak students, stores all results in the database, and updates the student's learning dashboard.

The data flow begins when the student interacts with the Flutter mobile application. Authentication requests are sent to Firebase Authentication for login validation. After successful login, the student profile is loaded from Cloud Firestore. When the learner performs a quiz or virtual experiment, the mobile application sends the activity data to the Python backend. The backend evaluates quiz answers, analyzes experiment outcomes, generates sub-lesson performance scores, detects weak areas, produces personalized feedback, and where weakness is identified, assigns a specific learning path for that student. Once the student completes the learning path and reattempts the quiz or experiment, the backend re-evaluates the new results. If the student still performs poorly, the backend assigns additional external resources for that specific weak sub-lesson. All processed results are stored in Cloud Firestore and retrieved by the dashboard and home page to display lesson-based analytics, feedback messages, learning path recommendations, and additional resource assignments, allowing the student to monitor progress and access escalating targeted support over time.

From a scalability perspective, the architecture is modular, which means each component can be extended independently without affecting the whole system. New lessons, experiments, quizzes,

learning path resources, additional external resources, and games can be added for additional topics or grade levels. Cloud Firestore supports scalable real-time data storage, while the backend can be expanded to handle more analytics features, learning path content, and escalating resource assignment logic as the user base grows. From a security perspective, the system uses secure authentication through Firebase, controlled database access, and authenticated API communication between the mobile application and backend. Student information is limited to necessary educational data, and performance records, learning path assignments, reattempt results, and additional resource assignments are stored securely to protect user privacy. This architecture therefore supports both future expansion and safe educational use.

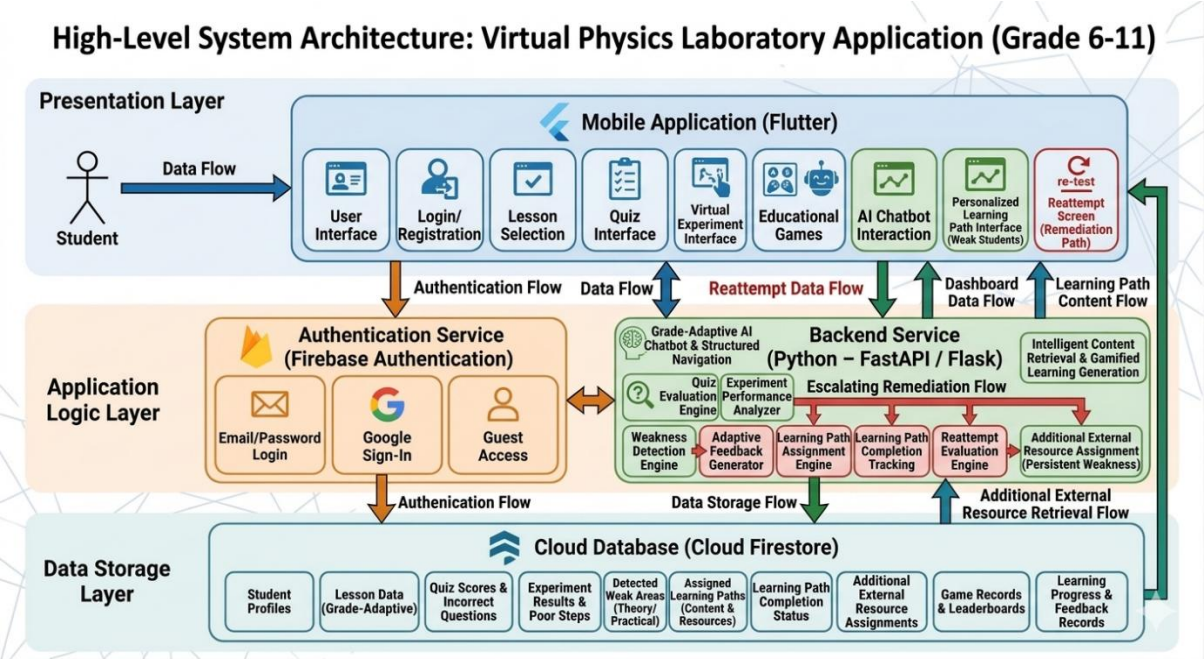


Figure 3-High-level system Archytecture

5. USER REQUIREMENTS

Stakeholder Identification

The proposed system involves several stakeholders who interact with or benefit from the mobile virtual physics laboratory application.

• **Students (Primary Users)** – Grade 6–11 students who use the mobile application to perform virtual physics experiments, answer quizzes, access learning materials, and monitor their learning progress..

• **School Administrators** – Educational institutions that may adopt the platform to improve access to laboratory-based learning for students.

• **Researchers / Developers** – Individuals responsible for developing, maintaining, and improving the system based on educational and technical requirements.

• **Educational Authorities** – Organizations responsible for curriculum development and digital education initiatives that may use the platform to support physics education in rural schools.

Functional Requirements

The system must provide the following functionalities.

1. The system shall allow students to register and create user accounts using email and passwords.
2. The system shall support secure login authentication, including Google Sign-In and guest access.
3. The system shall allow students to create and manage personal profiles, including name, grade level, and learning history.
4. The system should provide grade-based access to learning materials, including experiments, quizzes, and educational games.
5. The system shall allow students to perform virtual physics experiments through the mobile application.
6. The system shall record experiment completion status and evaluate practical performance based on task completion and accuracy of each experiment step.
7. The system shall provide intelligently generated sub-lesson quizzes covering Grade 6–11 curriculum content, including Multiple Choice Questions and structured questions to assess theoretical understanding at different cognitive levels.
8. The system shall automatically calculate and display separate quiz and experiment performance scores for each sub-lesson.
9. The system shall identify which quiz questions were answered incorrectly and which experiment steps were performed poorly for each sub-lesson.
10. The system shall detect weak theory and practical areas per sub-lesson based on quiz and experiment performance results.

11. The system shall generate personalized feedback messages and display improvement suggestions on the student home page based on detected weak areas.
12. The system shall assign a personalized learning path to weak students when low performance is detected in a specific sub-lesson, providing targeted resources to improve their knowledge in that weak area.
13. The system shall provide learning path resources including simple explanations, guided examples, correct experiment procedures, video-based learning materials, and external resources specific to the weak sub-lesson.
14. The system shall support repeated quiz attempts with shuffled question sets to encourage deeper conceptual understanding and prevent memorization-based answering.
15. The system shall allow students to reattempt the quiz and experiment for a weak sub-lesson after completing the assigned personalized learning path, to demonstrate improved understanding.
16. The system shall assign additional external resources such as YouTube videos and reference materials to students who still show low performance after completing the learning path and reattempting the quiz and experiment.
17. The system shall enforce structured learning progression by preventing students from advancing to the next sub-lesson without completing the current one.
18. The system shall store student learning data, including quiz scores, experiment outcomes, detected weak areas, assigned learning paths, and lesson progress.
19. The system shall display learning analytics and progress indicators through a student dashboard, including sub-lesson performance summaries and feedback messages.
20. The system shall allow students to track lesson-wise and sub-lesson-wise learning progress over time.
21. The system shall integrate with AI chatbot support for answering physics-related questions..

Non-Functional Requirements

The system must satisfy the following quality requirements.

Security

• The system shall implement secure authentication mechanisms for user login and registration.
• Student data shall be stored securely in the cloud database.
• Only authenticated users shall access personal learning data and dashboards.
• Communication between the mobile application and backend services shall be protected using secure APIs.

Performance

• The system shall provide fast response time for login, quiz evaluation, and data retrieval.
• The mobile application shall run efficiently on standard Android smartphones.
• Learning analytics data should be displayed without significant delays.

Usability

• The user interface shall be simple and intuitive for school students.
• Navigation between lessons, experiments, quizzes, and dashboards shall be easy to understand.
• The application shall provide clear feedback messages and learning progress indicators.

Reliability

• The system shall store learning progress data accurately.
• The system shall maintain stable operation during learning activities such as quizzes and experiments.

Scalability

• The system architecture shall support future expansion to additional lessons, experiments, and users.
• The cloud database shall allow scalable storage of student learning data.

COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the proposed system includes school students in Grades 6–11, particularly those studying physics in schools that have limited laboratory resources. The application is especially useful for students in rural and semi-urban educational environments where access to practical laboratory equipment is restricted. Secondary target users include teachers and educational institutions that wish to supplement classroom learning with digital experiments and interactive learning activities. The typical user persona is a school student who owns or has access to a smartphone and wants an interactive and accessible way to understand physics concepts and perform virtual experiments.

Value Proposition

The proposed system provides a mobile-based virtual physics learning environment that allows students to perform digital experiments, complete quizzes, and track their learning progress. Unlike traditional learning methods that rely heavily on theoretical explanations, the platform offers interactive and visual learning experiences that help students understand scientific concepts more effectively. The system also integrates learning analytics to help students monitor their progress and identify weak areas. This combination of experimentation, assessment, and progress tracking creates a more engaging and structured learning experience.

Revenue Model

The system can follow a freemium educational technology model. The basic version of the application can be provided free of charge to ensure accessibility for students, particularly in rural schools. Additional features such as advanced analytics, extended experiment libraries, premium educational content, and teacher monitoring tools could be offered through subscription-based services or institutional licenses. Educational institutions or private learning centers may subscribe to these premium features to enhance their teaching resources.

Cost Estimation

The development cost of the proposed system is relatively low because it uses open-source development frameworks and cloud services with free usage tiers. Technologies such as Flutter, Firebase Authentication, Cloud Firestore, and Python backend frameworks reduce the need for expensive software licenses. The primary costs would involve developer time, server resources for scaling the application, and maintenance activities if the system is deployed at a larger scale.

Pricing Strategy

A tiered pricing strategy can be adopted for the system. The core educational features, including basic experiments and quizzes, can remain free to encourage widespread adoption among students. Premium features such as additional experiments, advanced analytics dashboards, teacher monitoring tools, and expanded educational content can be offered through monthly or yearly subscription plans. Educational institutions may also receive discounted institutional packages that allow multiple students to access premium features.

Competitive Advantage

The proposed platform offers several advantages compared to existing digital learning systems. It combines virtual experiments, quizzes, learning analytics, and AI-assisted guidance within a single mobile application. Many existing educational tools provide only individual features such as simulations or quizzes. By integrating multiple learning components into a unified platform, the system provides a more comprehensive learning environment. Additionally, the mobile-based design makes the platform accessible to students who may not have access to traditional laboratory resources.

Intellectual Property Considerations

The system may involve intellectual property related to software architecture, experiment simulation designs, and learning analytics algorithms. These components can be protected through software copyrights and appropriate licensing agreements. If the system is expanded into a commercial educational product, the developers may consider registering trademarks for the platform name and protecting unique learning features to maintain ownership and prevent unauthorized duplication of the software.

7. BUDGET AND JUSTIFICATION

Hardware Costs

The proposed system does not require specialized hardware equipment because the application is designed to run on standard mobile devices and personal computers used for development. The development process can be carried out using existing computers available to the research team. Students who use the system will access the application through Android smartphones, which are widely available among school students. Therefore, no additional hardware purchases are required for the research prototype.

Software Licenses

The system will be developed using open-source software frameworks and free development tools, which eliminates the need for purchasing software licenses. Technologies such as Flutter for mobile development, Python frameworks (FastAPI or Flask) for backend services, and development environments like Visual Studio Code or Android Studio are freely available. Since these tools are open-source or provide free usage for development purposes, the project does not incur licensing expenses.

Cloud Services

The proposed system uses Firebase Authentication and Cloud Firestore for user authentication and data storage. These services provide free-tier usage plans that are sufficient for a research prototype and small-scale testing. The free tier allows secure user login management, storage of student profiles, quiz scores, and experiment performance data. If the system is deployed for large-scale use in the future, additional cloud resources may be required, but for the research stage, the available free cloud services are adequate.

Human Effort Cost

The system development is carried out as part of an undergraduate research project, and therefore the development work is performed by student researchers as part of their academic requirements. No external developers or paid technical staff are required during the research phase. Consequently, the human effort involved in designing, developing, and testing the system is considered part of the academic learning process and does not incur financial cost.

Justification for Each Expense

Overall, the project is designed as a research prototype, which allows the use of freely available development tools and cloud service tiers. By utilizing open-source frameworks, existing computing resources, and student development effort, the total estimated cost for system development remains minimal. This cost-efficient approach allows research to focus on evaluating the system’s educational impact and technical feasibility without requiring significant financial investment.

1.Development Tools

Table 1-development of tools cost

Tool	Purpose	Cost
Flutter	Mobile app development	$0
Android Studio / VS Code	Coding environment	$0
Python (FastAPI / Flask)	Backend development	$0
Git / GitHub	Version control	$0

2.Firebase Services

Table 2-Firebase services cost

Feature	Free Limit
Authentication (Email / Google login)	Free
Firestore Database	1 GB storage
Database reads	50,000 per day
Database writes	20,000 per day
Storage	Small free quota

3.Final Budget Estimation

Table 3-final budget

Component	Cost
Flutter Development Framework	$0
Python Backend (FastAPI / Flask)	$0
Firebase Authentication	$0
Cloud Firestore Database (1GB free tier)	$0
Google Sign-In Integration	$0
Development Tools (VS Code / Android Studio)	$0
Application Publishing	$0 (not publishing)

Total Estimated Budget = $0

8. WORK BREAKDOWN STRUCTURE (WBS)

Task Breakdown per Member

The project is divided into four main components, with each team member responsible for developing and integrating a specific module of the system.

Sachintha Nirmal S.M – AI Chatbot and Learning Navigation Module
Responsible for designing and developing the AI-powered chatbot that assists students in understanding physics concepts. This includes implementing grade-adaptive responses, guiding students through lesson content, and integrating the chatbot with the application's structured learning navigation system. The member also designs the hierarchical structure of grades, lessons, topics, experiments, and activities to ensure students can easily access learning materials.

Wijerathna C D W – Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module
Responsible for implementing the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module. This includes monitoring student quizzes and virtual experiment performance, analyzing learning progress at sub-lesson and topic levels, detecting weak areas, generating adaptive personalized feedback, providing remediation resources for weak students, and implementing structured learning progression enforcement within the mobile virtual physics laboratory.

Vidurangi – Interactive Physics Experimentation and Learning Module
Responsible for developing interactive physics simulations and virtual experiments within the mobile application. This includes designing practical experiment activities where students can interact with virtual objects, measure physical outcomes, and understand physics calculations through guided explanations.

Arthanayake C T H – Intelligent Content Retrieval and Gamified Learning Generation Module
Responsible for implementing the intelligent content retrieval system and gamified learning features. This includes developing keyword-based search functionality for physics learning materials and creating 2D educational games aligned with lesson topics to improve student engagement and motivation.

Timeline Alignment

The project development process is organized into several phases aligned with the research timeline.

• **Phase 1 – Requirement Analysis and Literature Review**
Identification of research problems, review of existing studies, and definition of system requirements.

• **Phase 2 – System Design**
Design of system architecture, database structure, user interface layouts, and component interactions.

• **Phase 3 – System Development**
Implementation of each module by the respective team members, including mobile interface development, backend services, experiment simulations, and AI chatbot integration.

• **Phase 4 – Integration and Testing**
Integration of all system components, testing system functionality, verifying data flow between modules, and fixing identified issues.

• **Phase 5 – Evaluation and Documentation**
Evaluation of system performance, preparation of research documentation, and final report submission.

Balanced Workload Distribution

The project tasks are distributed among team members to ensure a balanced workload and effective collaboration. Each member is responsible for a major system component while contributing to system integration and testing activities. This modular development approach allows parallel progress in different parts of the system while maintaining coordination among team members. By assigning clearly defined responsibilities, the team ensures efficient development, effective knowledge sharing, and timely completion of the research project.

WBS Table

Table 4-WBS Table

WBS ID	Task Name	Member	Description	Duration
1.0	Project Initiation	All	Identify research problems, define project scope, and obtain supervisor approval	Month 1
1.1	Literature Review	All	Review research papers related to virtual labs, mobile learning, learning analytics, adaptive feedback, RAG-based chatbots, gamification, and personalized learning systems	Month 1 – Month 2
1.2	Research Proposal Preparation	All	Prepare research proposal including introduction, research gaps, objectives, and methodology for all four modules	Month 2
2.0	System Requirement Analysis	All	Identify functional and non-functional requirements of the complete mobile virtual physics laboratory system	Month 3
2.1	System Architecture Design	All	Design overall system architecture including Flutter mobile application, Python backend services, Firebase, and database structure	Month 3
2.2	Database Design	All	Design Cloud Firestore data structure for user profiles, quiz scores, experiment data, analytics records, chatbot data, game records, learning paths, and curriculum content	Month 3
3.0	User Authentication	Wijerathna C D W	Implement student registration, login, Google Sign-In, guest access, and student profile management using	Month 4

	and Profile Module		Firebase Authentication and Cloud Firestore	
3.1	Intelligent Quiz Evaluation System	Wijerathna C D W	Develop intelligently generated sub-lesson quizzes with MCQs and structured questions aligned to Grade 6–11 curriculum, with shuffled retake support	Month 5 – Month 6
3.2	Experiment Performance Evaluation	Wijerathna C D W	Implement system to evaluate practical experiment steps, record performance outcomes, and identify poorly performed experiment areas per sub-lesson	Month 6 – Month 7
3.3	Learning Analytics and Weakness Detection	Wijerathna C D W	Develop algorithms to analyze sub-lesson quiz and experiment results and detect weak theory and practical areas	Month 7 – Month 8
3.4	Adaptive Feedback Generation	Wijerathna C D W	Implement rule-based feedback mechanism to generate personalized feedback messages and display improvement suggestions on the student home page	Month 8
3.5	Personalized Learning Path Assignment	Wijerathna C D W	Develop learning path assignment system that detects weak students and assigns targeted resources including explanations, guided examples, experiment procedures, and video materials; enable reattempt after completion; assign additional external resources for persistent weakness	Month 8 – Month 9
3.6	Dashboard and Visualization Development	Wijerathna C D W	Develop student dashboard with sub-lesson performance indicators, quiz and experiment scores, weak area summaries, feedback messages, and learning path access	Month 9 – Month 10
4.0	Grade-Tagged Physics Knowledge Base	Sachintha Nirmal S.M	Build a grade-tagged physics knowledge base containing curriculum-aligned physics content organized by grade level, lesson, topic, and activity for RAG retrieval	Month 4 – Month 5
4.1	RAG-Based AI Chatbot Pipeline	Sachintha Nirmal S.M	Implement the Retrieval-Augmented Generation pipeline using LLM to retrieve relevant syllabus content and	Month 5 – Month 6

			generate physics explanations for student queries	
4.2	Grade-Adaptive Response Generation	Sachintha Nirmal S.M	Develop grade-adaptive response generation that tailors chatbot answers based on the student's grade level and lesson context	Month 6
4.3	Structured Learning Workflow Module	Sachintha Nirmal S.M	Design and implement the structured learning workflow including hierarchical navigation for grades, lessons, topics, experiments, and activities	Month 4 – Month 5
4.4	Chatbot Integration with Mobile Frontend	Sachintha Nirmal S.M	Integrate the AI chatbot interface and structured learning workflow screens into the Flutter mobile application	Month 7
5.0	Syllabus-Aware Keyword Search Engine	Arthanayake C T H	Implement keyword-based search engine that matches student-entered physics keywords with Sri Lankan National curriculum syllabus materials	Month 4 – Month 6
5.1	Adaptive 2D Game Generation Engine	Arthanayake C T H	Create curriculum-aligned 2D educational games using Unity that adapt game difficulty and content based on student grade level and lesson topic	Month 5 – Month 7
5.2	Grade-Adaptive Scoping Framework	Arthanayake C T H	Design scoping framework that maps digital learning activities and game difficulty to structured lesson-level objectives for Grades 6–11	Month 6 – Month 7
5.3	Content Retrieval and Game Integration	Arthanayake C T H	Integrate keyword-based retrieval system and game generation module with Firebase Firestore for NIE-aligned curriculum data storage and retrieval	Month 7 – Month 8
6.0	Virtual Experiment Simulation Development	Vidurangi H I D G	Develop interactive physics virtual experiments using Unity 2D where students can interact with virtual objects and perform guided experiments	Month 4 – Month 7
6.1	Measurement Extraction Module	Vidurangi H I D G	Implement real-time measurement extraction system using OpenCV and YOLOv8 to capture physics measurements such as distance, time, and velocity from student interactions	Month 5 – Month 6
6.2	Physics Calculation Module	Vidurangi H I D G	Apply physics formulas to extracted measurements to generate step-by-step	Month 6 – Month 7

			calculation results and concept explanations for students	
6.3	Experiment Performance Evaluation Module	Vidurangi H I D G	Implement scoring system that evaluates student experiment performance based on safety tool usage, apparatus selection, procedure steps, and final result accuracy	Month 7 – Month 8
7.0	System Integration	All	Integrate Flutter mobile application, Python backend analytics services, Firebase Authentication, Cloud Firestore, chatbot, experiment module, gamified learning, and analytics module into a complete unified system	Month 10 – Month 11
8.0	System Testing and Debugging	All	Conduct functional and performance testing of all modules including authentication, quiz evaluation, experiment evaluation, weakness detection, feedback generation, learning path assignment, chatbot accuracy, RAG retrieval, experiment simulations, and game functionality	Month 10 – Month 11
9.0	System Evaluation	All	Evaluate usability and effectiveness of the complete system including mastery level prediction accuracy, chatbot performance, RAG retrieval accuracy, experiment evaluation accuracy, search relevance, game engagement, and learning path effectiveness	Month 11
10.0	Documentation and Final Report	All	Prepare final research documentation including results, discussion, and complete system description for all four modules	Month 11 – Month 12
11.0	Final Presentation Preparation	All	Prepare presentation slides, demonstration materials, and final system submission	Month 12

9. GANTT CHART

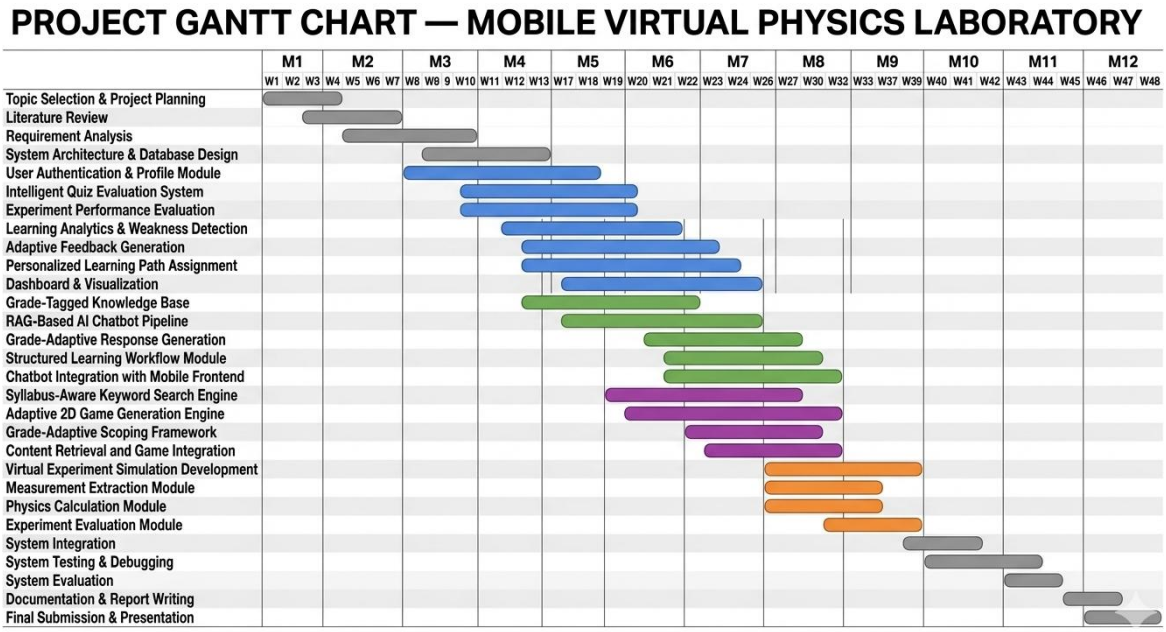


Figure 4-Gantt chart

10. REFERENCES LIST

[1] R. Elmoazen, M. Al-Mamun and A. Al-Hassan, "Learning Analytics in Virtual Laboratories: A Systematic Literature Review of Empirical Research," *Smart Learning Environments,* vol. 10, no. 1, p. 20, 2023.

[2] H. Matovu, D. A. Kencana Ungu, M. Won, C.-C. Tsai, D. F. Treagust, M. Mocerino and R. Tasker, "Immersive Virtual Reality and Learning: A Meta-Analysis," *Educational Psychology Review,* vol. 59, no. 2, p. 1–41, 2021.

[3] G. and A. Aljanazrah, "The Effectiveness of Using Virtual Experiments on Students' Learning in the General Physics Lab," *Journal of Information Technology Education: Research,* vol. 15, no. 11, pp. 1-20, 2020.

[4] M. Ally, Mobile Learning: Transforming the Delivery of Education and Training, Athabasca, Alberta, Canada: Athabasca University Press, 2009.

[5] M. Kalogiannakis, S. Papadakis and M. Zourmpakis, "Gamification in Science Education: A Systematic Review of the Literature," *Education Sciences,* vol. 11, no. 1, pp. 1–36, 2021.

[6] G. Siemens and P. Long, "Penetrating the Fog: Analytics in Learning and Education," *EDUCAUSE Review,* vol. 46, no. 5, pp. 1-6, 2011.

[7] K. Sukhija, M. Jindal and N. Aggarwal, "The Recent State of Educational Data Mining: A Survey and Future Visions," Amritsar, India, 2015.

[8] R. Ferguson, "Learning Analytics: Drivers, Developments and Challenges," *International Journal of Technology Enhanced Learning,* vol. 4, no. 5/6, p. 1–14, 2013.

[9] G.-J. Hwang and S.-Y. Fu, "Advancement and Research Challenges of Personalised Learning with Big Data and Artificial Intelligence: A Literature Review," *International Journal of Mobile Learning and Organisation,* vol. 14, no. 1, p. 58–84, 2020.

[10] O. Zawacki-Richter, V. Marín, M. Bond and F. Gouverneur, "Systematic Review of Research on Artificial Intelligence Applications in Higher Education," *International Journal of Educational Technology in Higher Education,* vol. 16, no. 1, pp. 1-27, 2019.

[11] Z. C. Zacharia and C. P. Constantinou, "The Use of Computer-Based Simulations in Physics Teaching and Learning," *Computers & Education (Elsevier),* vol. 51, no. 1, 2008.

[12] C. E. Wieman, W. K. Adams and K. K. Perkins, "PhET Interactive Simulations: Research-Based Simulations for Physics Education," *American Journal of Physics,* vol. 76, no. 4, pp. 1-12, 2008.

[13] R. F. Bodily and K. Verbert, "Learning Analytics Dashboards: A Systematic Literature Review," *IEEE Transactions on Learning Technologies,* vol. 10, no. 4, p. 389–399, 2017.

[14] A. Coban, D. Dzsotjan, S. Küchemann, J. Durst and J. Kuhn, "AI Support Meets AR Visualization for Alice and Bob: Personalized Learning Based on Individual ChatGPT Feedback in an AR Quantum Cryptography Experiment for Physics Lab Courses," arXiv, Ithaca, New York, USA, 2024.

[15] F. Ouyang, M. Wu, L. Zheng, L. Zhang and P. Jiao, "Integration of Artificial Intelligence Performance Prediction and Learning Analytics to Improve Student Learning in Online Engineering Course," *International Journal of Educational Technology in Higher Education,* vol. 20, no. 4, pp. 1-23, 2023.

[16] C. Halkiopoulos and E. Gkintoni, "Leveraging AI in E-Learning: Personalized Learning and Adaptive Assessment Through Cognitive Neuropsychology—A Systematic Analysis," *Electronics,* vol. 13, no. 18, p. 3762, 2024.

[17] A. Grimm, A. Steegh, M. Kubsch and K. Neumann, "Learning Analytics in Physics Education: Equity-Focused Decision-Making Lacks Guidance!," *Journal of Learning Analytics,* vol. 10, no. 1, pp. 71-84, 2023.

11.APPENDICES

[Appendix A — Survey instruments]

This research does not include survey-based data collection methods at this stage. Instead, the system evaluation is conducted through functional testing, user interaction testing, and performance analysis using the System Functionality Checklist. Therefore, no survey instruments are included in this study.

[Appendix B — Detailed Diagrams]

This section includes detailed diagrams related to the system design and methodology of the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module. These diagrams provide a visual representation of the research workflow, module structure, and system data flow.

Key diagrams include:

- **Research Methodology Diagram** — illustrates the overall design science research approach used to develop the module
- **Research Methodology Process Diagram** — shows the step-by-step process flow from student login through quiz and experiment assessment, weakness detection, learning path assignment, reattempt, and escalating resource provision
- **High-Level System Architecture Diagram** — presents the three-layer architecture including the Flutter presentation layer, Python application logic layer, and Cloud Firestore data layer with all module components and their interactions
- **Module Architecture Diagram** — illustrates the five main parts of the module: user authentication, profile management, sub-lesson assessment, learning analytics and weakness detection, and personalized learning path assignment
-

These diagrams illustrate the flow of activities, system components, and interactions between the mobile application, Python backend, Firebase Authentication, Cloud Firestore database, and external resources such as YouTube videos and reference materials.

[Appendix C — Preliminary UI Mockups]

- Student login and registration screen
- Home page screen with feedback messages and suggestion section
- Sub-lesson selection screen showing lesson and sub-lesson structure
- Quiz attempt screen with MCQ and structured question layout
- Quiz result screen displaying quiz score and identified weak questions
- Virtual experiment screen with step-by-step guided interaction
- Experiment result screen displaying experiment score and weak step identification
- Student dashboard screen showing sub-lesson performance indicators and weak area summaries
- Personalized learning path screen showing assigned resources for weak sub-lessons including explanations, guided examples, experiment procedures, and video materials



Figure 5-UI sample mockups

[Appendix D — Risk Analysis]

Table 5-Risk analysis

Risk	Likelihood	Impact	Mitigation Strategy
Inaccurate weakness detection due to insufficient quiz and experiment data	Medium	High	Validate detection logic with sample student data during testing phase
Students bypassing learning path by not engaging with assigned resources	Medium	Medium	Enforce structured progression so students cannot advance without completing the current sub-lesson
Firebase free tier limits exceeded during testing with multiple users	Low	Medium	Limit concurrent test users during prototype evaluation stage
Video-based learning materials unavailable due to broken YouTube links	Medium	Medium	Use multiple backup links and verify all links before system evaluation
Python backend response delays affecting real-time feedback display	Low	High	Optimize analytics processing logic and test backend performance under load
Student quiz data loss due to network interruption	Low	High	Implement local caching in Flutter app and sync with Cloud Firestore on reconnection
Shuffled quiz retakes generating repeated questions	Low	Medium	Implement question bank with sufficient variation per sub-lesson to ensure unique sets

[Appendix E — Ethical Documents]

- Informed consent form for student participants involved in system usability testing
- Parental consent form for underage student participants (Grade 6–11)
- Data privacy and confidentiality agreement ensuring student learning data is used solely for research purposes
- Data handling policy outlining how student profiles, quiz scores, experiment results, and learning path records are stored, accessed, and protected in Cloud Firestore
- The system is designed to support fair and inclusive access to education without bias

Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students

Project ID - R26-IT-150

Project Proposal Report

VIDURANGI H I D G – IT22143990

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Vidurangi H I D G	IT22143990	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Indeewari Dekeiwatte	18/03/2026	

ABSTRACT

Physics education is essential for developing scientific thinking and problem-solving skills among school students. However, many learners struggle to understand physics concepts due to the abstract nature of theoretical explanations and the lack of practical experimentation opportunities. This issue is particularly significant in rural schools where laboratory facilities and learning resources are limited. As a result, students often learn physics mainly through textbooks and lectures, which reduces engagement and makes it difficult to connect theoretical formulas with real-world observations.

Although several digital learning platforms and simulations have been introduced to support science education, most existing systems focus only on single learning approaches such as simulations, quizzes, or gamified content. These tools often lack integration between interactive activities, measurable experimental results, and step-by-step explanations of physics calculations. Consequently, students may interact with simulations or games without fully understanding the mathematical relationships behind the observed outcomes.

To address this limitation, this research proposes the development of an Interactive Physics Experimentation and Learning Module designed for Grade 6–11 students, particularly in rural educational environments. The proposed system integrates physics-based games, real-time measurement generation, step-by-step calculation explanations, and a virtual practical laboratory environment within a mobile application. In this module, students can interact with physics activities such as motion-based simulations where their actions generate measurable data (e.g., distance travelled or motion). The system then explains the relevant physics formulas and calculations to help learners understand how theoretical concepts are applied in practice. Additionally, the virtual laboratory allows students to conduct digital experiments using simulated laboratory tools.

The system will be developed using mobile application technologies, interactive simulation frameworks, and physics-based algorithms for measurement calculation and visualization. The expected outcomes include improved conceptual understanding of physics, increased student engagement, and greater accessibility to experimental learning for students who lack laboratory facilities.

From a social perspective, the system supports inclusive and equitable quality education, particularly benefiting rural learners. Commercially, the platform has potential for adoption by schools and educational institutions as an affordable digital learning tool.

Keywords: Physics Education, Game-Based Learning, Virtual Laboratory, Interactive Simulations, Mobile Learning.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor for their continuous guidance, encouragement, and valuable feedback throughout the development of this research proposal. Their support and expertise helped me to clearly understand the research problem and improve the overall quality of this work.

I am also thankful to the academic staff of the Department of Information Technology at the Sri Lanka Institute of Information Technology for providing the knowledge, academic resources, and learning environment necessary to carry out this research study. Their teaching and support have greatly contributed to my academic development.

In addition, I would like to acknowledge my fellow project team members for their cooperation, discussions, and support during the early stages of this research work. Collaborative discussions helped in shaping the overall concept of the proposed system.

Finally, I would like to express my deepest appreciation to my family and friends for their constant encouragement, patience, and motivation throughout my academic journey. Their support has been invaluable in helping me successfully complete this research proposal.

Table of Contents

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
LIST OF TABLES 6
LIST OF FIGURES 7
LIST OF ABBREVIATIONS 8
LIST OF APPENDICES 9
1 INTRODUCTION 10
1.1 Background and Context 10
1.2 Literature Review 11
1.3 Research Gap 13
2. OBJECTIVE 14
2.1 Main Objective 14
2.2 Specific Objectives 14
3. METHODOLOGY 15
4. HIGH-LEVEL SYSTEM ARCHITECTURE 18
5. USER REQUIREMENTS 20
6. COMMERCIALIZATION PLAN 21
7. BUDGET AND JUSTIFICATION 23
8. WORK BREAKDOWN STRUCTURE (WBS) 25
9. GANTT CHART 30
10. References List 31
11.APPENDICES 32
APPENDIX 1 Survey instruments 32
APPENDIX 2 Detailed diagrams 32
APPENDIX 3 Preliminary UI mockups 32
APPENDIX 4 Risk analysis 33
APPENDIX 5 Ethical documents 34

LIST OF TABLES

Table 1 Budget and Budget Justification **24**
Table 2 Gantt Chart **30**
Table 3 Risk Analysis **33**

LIST OF FIGURES

Figure 1 Research Methodology Diagram **16**
Figure 2 Research Methodology Process Diagram **17**
Figure 03 High Level System Architecture Diagram **19**
Figure 04 Preliminary UI mockups **33**

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CV	Computer Vision
YOLO	You Only Look Once
YOLOv8	You Only Look Once Version 8(Object Detection Model)
UI	User Interface
UX	User Experience
DSR	Design Science Research
API	Application Programming Interface
DB	Database
SDG	Sustainable Development Goal
ICT	Information and Communication Technology
STEM	Science, Technology, Engineering, and Mathematics
FPS	Frames Per Second

LIST OF APPENDICES

APPENDIX 1 Survey instruments **32**
APPENDIX 2 Detailed diagrams **32**
APPENDIX 3 Preliminary UI mockups **32**
APPENDIX 4 Risk analysis **33**
APPENDIX 5 Ethical documents **34**

1 INTRODUCTION

1.1 Background and Context

Physics education continues to face significant challenges due to students' low engagement, poor conceptual understanding, and limited access to practical laboratory experiences. Many learners perceive physics as difficult, abstract, and disconnected from real-life applications, which negatively impacts motivation and academic performance [1] This issue affects secondary school students globally, particularly in developing countries, where access to laboratory facilities and modern learning tools is limited, making it difficult to achieve effective STEM education outcomes [2].For instance, international assessments such as PISA highlight low student performance in science subjects, emphasizing the urgency for improved teaching methodologies [1].

Existing solutions such as virtual laboratories and simulation tools have been introduced to address these challenges. Virtual labs enable students to perform experiments remotely and support observation, measurement, and analysis processes [3]. Similarly, interactive simulations like PhET have demonstrated improvements in student understanding, engagement, and academic performance [4] [5]. However, most of these systems focus mainly on visualization and conceptual learning rather than integrating real-time measurement generation, calculation processes, and structured experiment evaluation. Additionally, many systems operate as standalone tools and do not provide a comprehensive learning experience that combines gameplay, experimentation, and automated feedback in a single platform [3] [5].

To overcome these limitations, this research proposes an Interactive Physics Experimentation and Learning Module that integrates game-based learning with virtual experiments. The system generates real-time measurements (e.g., distance, time, velocity) from student interactions, applies physics formulas, and provides step-by-step calculation explanations. It also evaluates student performance based on apparatus selection, safety practices, procedures, and result accuracy. The system is developed using Unity 2D, OpenCV, YOLOv8, Flutter, Python, and Firebase. This solution supports Sustainable Development Goal 4 (Quality Education) by improving access to interactive and practical learning. Furthermore, it aligns with the AI-driven Educational Technology research cluster, promoting innovative, scalable, and student-centered digital learning solutions.

1.2 Literature Review

Recent research in physics education shows a growing shift from traditional lecture-based teaching toward interactive digital learning approaches such as simulations, virtual laboratories, and game-based learning. A major research approach in the literature is the use of interactive simulations to improve students' conceptual understanding, motivation, and academic performance. Studies based on PhET simulations report that simulation-based learning helps students visualize abstract concepts, engage actively with content, and achieve better learning outcomes than conventional teaching methods [1] [6] [7] [8]. These studies also show that simulation-supported learning is especially useful in secondary school physics, where many learners struggle with difficult topics and low motivation [1] [8].

Another important research approach is the use of computer simulations as substitutes or supplements for laboratory activities. Foundational studies found that well-designed simulations can support conceptual understanding and even help students perform better than those using only traditional equipment [2] [3]. Similarly, virtual laboratory research highlights that digital labs can support observation, measurement, data collection, analysis, and conclusion drawing when real laboratory access is limited [9]. These approaches are valuable for physics education because they reduce dependence on physical resources and provide more accessible practical learning opportunities.

A third stream of literature focuses on game-based learning and gamification. Research shows that educational games can make science learning more relevant, engaging, and enjoyable, while also improving content understanding [4]. Studies on gamification further suggest that game elements such as points, badges, challenges, and progression can increase learner motivation and participation [5] [10]. In addition, recent work has explored PhET not only as a simulation tool but also as a game-based learning environment, reporting improvements in both understanding and learner interest [6]. These findings are relevant to your component because they support the use of interactive and game-like environments in physics learning.

In terms of existing products and services, the reviewed studies mainly discuss tools such as PhET interactive simulations**,** virtual laboratory systems**,** and educational physics games like *Physicus* [2] [4] [9] [6]. PhET is the most frequently highlighted product because it is interactive, visually engaging, and widely used in school science education [1] [2] [6] [7] [8] Virtual lab systems are presented as useful alternatives when schools lack sufficient laboratory facilities [9], while physics games are discussed as tools that increase enjoyment and help students learn specific concepts through exploration and problem-solving [4]

However, the literature also reveals several limitations in existing systems. First, most simulation-based tools mainly support conceptual visualization and basic interaction, but they do not generate real-time measurable outputs such as distance, time, or velocity from student actions [1] [6] [8]. Second, many systems improve engagement but do not connect student interaction to automatic physics formula application and step-by-step calculation explanations [4] [10] Third, virtual labs often support observation and measurement, yet they rarely provide structured evaluation of student performance based on apparatus selection, safety tools, procedure steps, and final result accuracy [9]. Fourth, many existing tools function as separate

systems—simulations, games, tutoring environments, and virtual labs are usually designed independently rather than integrated into one learning platform [3] [9] [10] Finally, reviews of gamification indicate that although game elements can increase motivation, there is still limited evidence on how to design them effectively for long-term educational value [5] [10].

Overall, the literature confirms that simulations, virtual labs, and game-based learning can enhance physics education, but current systems remain fragmented. There is still limited research on an integrated mobile learning module that combines game-based interaction, real-time measurement generation, automatic calculation support, and virtual experiment evaluation within one platform. This gap provides the basis for the proposed Interactive Physics Experimentation and Learning Module**.**

1.3 Research Gap

Although existing studies confirm that simulations, virtual laboratories, and game-based learning can improve physics education, an important gap still remains in how these approaches are designed and integrated. Current systems mainly focus on concept visualization, motivation, or isolated virtual practice, but they do not provide a unified learning process that connects student interaction, measurable outputs, calculation reasoning, and experiment assessment in one platform [1] [3] [9] [6]. For example, PhET and similar interactive simulations are effective for improving understanding and engagement, yet they are mainly designed for exploration and conceptual learning rather than for generating real-time physics measurements from gameplay and converting them into structured calculation explanations [1] [2] [6] [8].

Likewise, virtual laboratory systems support observation, measurement, and data analysis, but they often function only as digital substitutes for experiments and do not evaluate students in a detailed manner based on apparatus selection, safety tools, procedural steps, and result accuracy [9]. Game-based learning studies show that educational games can enhance motivation, enjoyment, and content learning, but most of these systems stop at engagement and do not connect gameplay to physics formula application or measurable academic outputs such as distance, time, and velocity [4] [5] [10]. In addition, many existing approaches are implemented as separate tools—simulations for visualization, games for engagement, and virtual labs for experimentation—without offering an integrated mobile-based environment that combines all these learning functions [3] [9] [10].

Therefore, the key research gap is the absence of a single intelligent physics learning module that integrates game-based interaction, real-time measurement generation, automatic step-by-step calculation support, and structured virtual experiment evaluation. This makes current approaches insufficient for supporting both conceptual learning and practical skill development in a complete and continuous way [4] [9] [6]. The novelty of this research lies in proposing an Interactive Physics Experimentation and Learning Module that combines these missing elements in one mobile platform using Unity 2D, OpenCV, YOLOv8, Flutter, Python, and Firebase. This integrated design is creative because it moves beyond simple simulations or gamification and introduces a system where student actions directly produce measurable values, trigger formula-based reasoning, and support practical experiment assessment. Hence, this study contributes a more comprehensive and innovative solution to the current limitations in digital physics education.

2. OBJECTIVE

2.1 Main Objective

To develop and evaluate an interactive physics learning system that integrates game-based measurement, automated calculation, and virtual experimentation to improve conceptual understanding and practical skills of Grade 6–11 students.

2.2 Specific Objectives

- To capture real-time physics measurements (e.g., distance, time, velocity) from student interactions using object detection techniques.
- To apply physics formulas automatically and generate step-by-step calculation explanations based on measured values.
- To design and implement virtual laboratory experiments aligned with school physics curriculum using interactive digital tools.
- To evaluate student experiment performance based on apparatus selection, safety practices, procedural steps, and accuracy of results.
- To develop a mobile-based application that integrates games, calculations, and experiments into a user-friendly learning platform.
- To assess the effectiveness of the system in improving student engagement and understanding of physics concepts through user testing and performance analysis.

3. METHODOLOGY

Research Approach

This research follows a Design Science Research (DSR) approach combined with experimental evaluation. The Design Science approach is used to design and develop an innovative interactive physics learning system that integrates games, measurement, and virtual experiments. The experimental approach is applied to evaluate the effectiveness of the system in improving students' understanding, engagement, and performance in physics learning. This combination ensures both system development and validation in a real educational context.

Technical Methods and Algorithms

The system uses several technical methods to achieve its functionality. Interactive physics games and virtual experiments are developed using simulation techniques, where student actions generate measurable values such as distance, time, and velocity. Computer vision methods using OpenCV and YOLOv8 are applied for object detection and motion tracking to extract these measurements from simulation environments. A rule-based calculation engine is implemented to apply physics formulas and generate step-by-step explanations. Additionally, a performance evaluation algorithm is designed to assess experiments based on criteria such as apparatus selection, procedural correctness, safety considerations, and result accuracy.

Tools and Platforms with Justification

The system is developed using a combination of modern tools and platforms. Unity 2D is used to create interactive physics games and virtual experiments due to its strong support for real-time simulation and user interaction. OpenCV and YOLOv8 are used for object detection and measurement extraction because of their accuracy and efficiency in handling motion-related data. Flutter is selected for mobile application development as it enables cross-platform compatibility and a user-friendly interface**.** Python is used for backend processing and implementing calculation and evaluation logic due to its flexibility and strong library support. Firebase is used for cloud storage and real-time data management, ensuring scalability and accessibility.

Data Collection and Ethical Considerations

The system collects data related to student interactions, including generated measurements, calculation results, and experiment performance. This data is used to evaluate learning outcomes and system effectiveness. Ethical considerations are maintained by ensuring that no sensitive personal information is collected, and all data is used strictly for educational and research purposes. User consent is obtained where necessary, and data privacy and confidentiality are ensured through secure storage mechanisms.

Validation Strategy and Performance Metrics

The system is validated through user testing with students in the target group (Grade 6–11). The evaluation focuses on both system performance and learning outcomes. Key performance metrics include accuracy of measurement extraction**,** correctness of generated calculations, and reliability of experiment evaluation. Educational effectiveness is measured through improvements in student understanding, engagement levels, and task completion rates. User feedback is also collected to assess usability and overall learning experience.

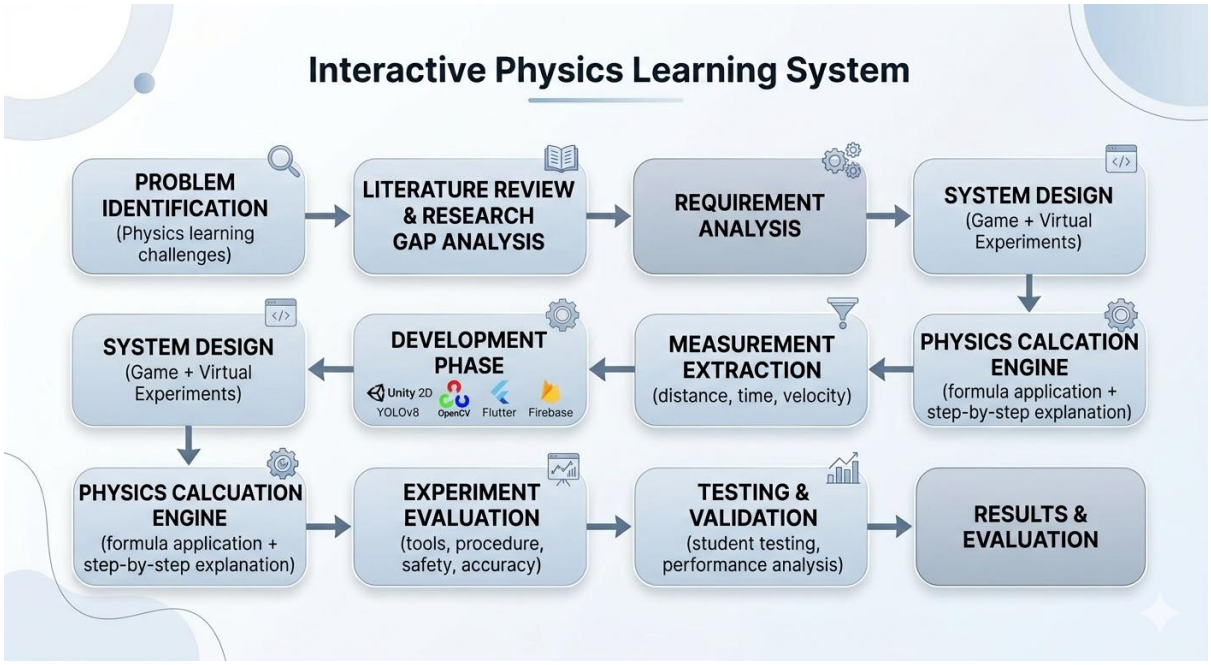


Figure 1 Research Methodology Diagram

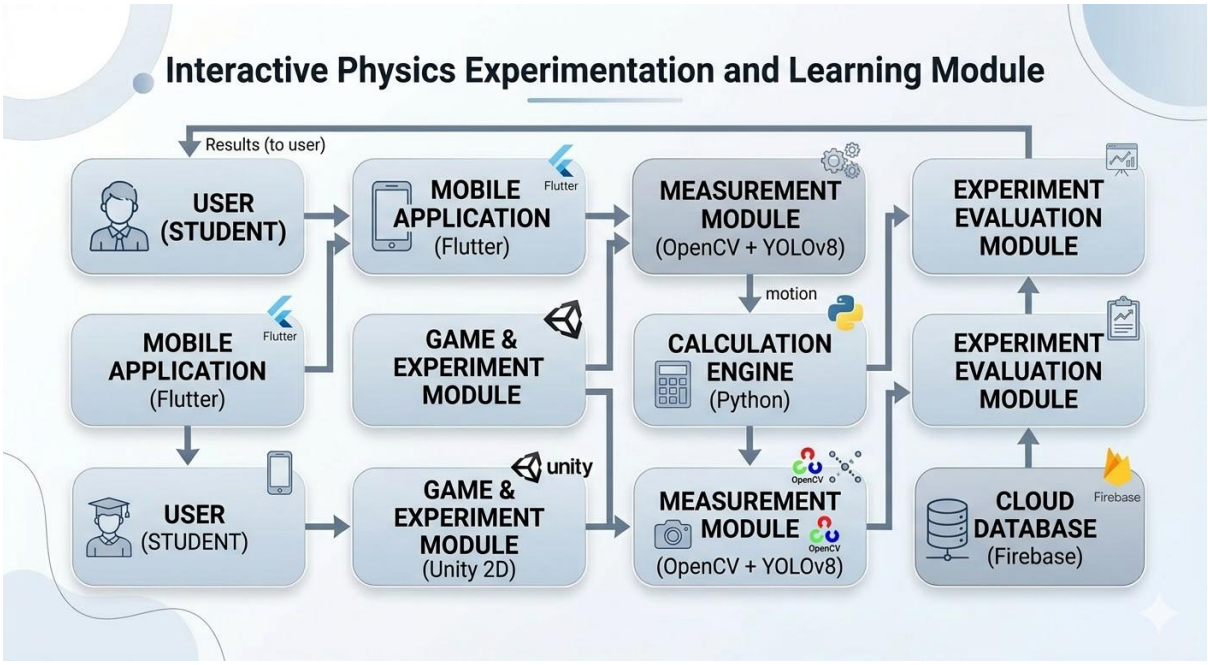


Figure 2 Research Methodology Process Diagram

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed solution is a mobile-based interactive physics learning system designed to support students in understanding physics concepts through games, measurements, calculations, and virtual experiments. The high-level architecture consists of several connected components that work together to provide a complete learning experience. The main components are the mobile application interface**,** game and experiment module**,** measurement extraction module**,** calculation engine**,** experiment evaluation module**,** backend services, and cloud database. The mobile application, developed using Flutter, acts as the main access point for students. Through this interface, users can select physics games, perform virtual experiments, view generated measurements, and receive feedback. The game and experiment modul**e**, developed using Unity 2D, contains interactive simulations where students perform actions that produce measurable outputs such as distance, time, and velocity. This is the core individual component of the system, and it consists of two parts: the physics game section and the virtual experiment section.

The measurement extraction module uses OpenCV and YOLOv8 to identify objects, track movement, and capture measurement-related values from the simulations. These values are passed to the calculation engine, implemented in Python, where physics formulas are applied to generate results and step-by-step explanations. In parallel, the experiment evaluation module analyzes student performance in virtual practical tasks based on apparatus selection, safety tools, procedural steps, and final result accuracy. All processed data, user progress, and results are managed by backend services and stored in Firebase for real-time access and synchronization.

The interaction between components follows a continuous flow. First, the user interacts with a game or experiment through the mobile app. The game environment sends simulation data to the measurement extraction module. After extracting the necessary values, the system forwards them to the calculation and evaluation modules. The generated outputs are then returned to the mobile app and displayed to the student as measurement results, calculation steps, and experiment feedback. This data flow ensures that student interaction is directly transformed into meaningful academic guidance.

From a scalability perspective, Firebase supports real-time cloud storage and multiple concurrent users, making the system suitable for large school deployments. In terms of security, user data is stored securely with controlled access, and only necessary learning-related information is collected to protect privacy and confidentiality.

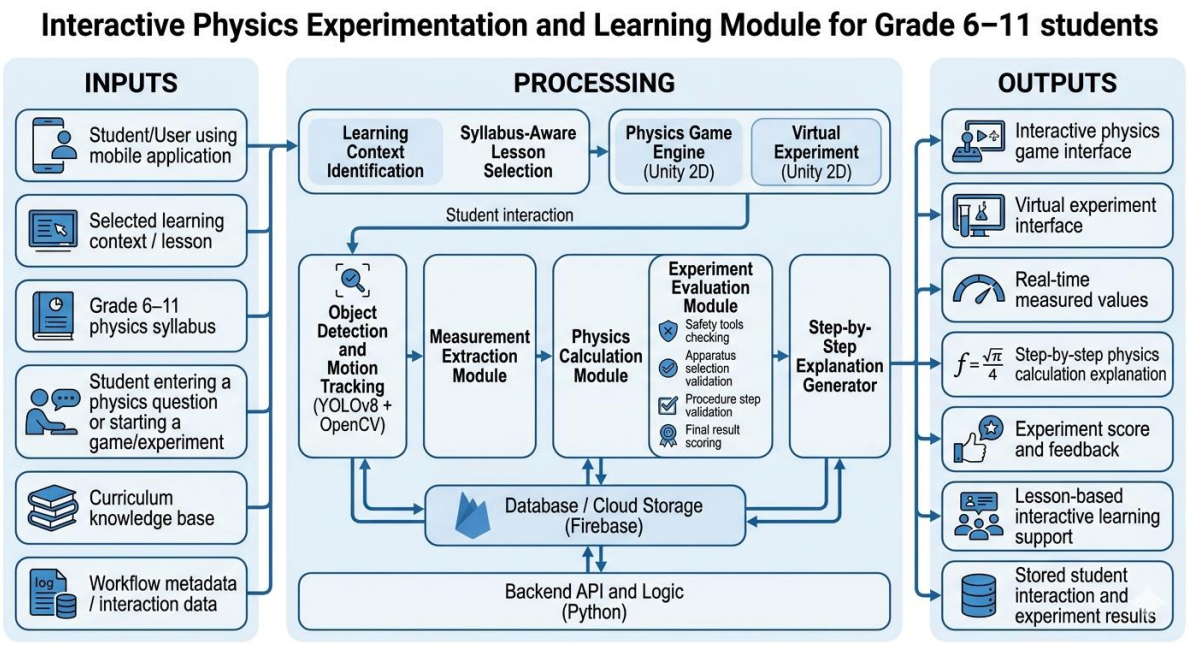


Figure 03 High Level System Architecture Diagram

5. USER REQUIREMENTS

- Stakeholders
  1. **Students**– Grade 6–11 students who use the mobile application to learn physics concepts through interactive games and virtual experiments.
  2. **School Administrators** – Responsible for supporting the adoption of the system in schools and encouraging its use for educational purposes.
  3. **Developers / Researchers** – Responsible for designing, developing, maintaining, and improving the learning module.
  4. **Educational Institutions** – Schools or educational organizations that may adopt the system to enhance digital learning and support physics education.

- Functional requirements
  - The system shall allow users to play interactive physics-based games.
  - The system shall capture real-time measurements such as distance, time, and velocity from user interactions.
  - The system shall extract measurement data using object detection and motion tracking techniques.
  - The system shall apply physics formulas automatically based on generated values.
  - The system shall display step-by-step calculation explanations to enhance understanding.
  - The system shall allow users to perform virtual laboratory experiments using digital tools.
  - The system shall evaluate experiment performance based on apparatus selection, safety, procedural steps, and accuracy of results.
  - The system shall store user data and progress in a cloud database.
  - The system shall provide feedback and results to users after each activity.
  - The system shall support mobile-based access through a user-friendly interface

- Non-functional requirements
  - **Performance:** The system should provide fast response times and real-time interaction without delays.
  - **Usability:** The interface should be simple, intuitive, and suitable for school students.
  - **Security:** User data must be securely stored, and unauthorized access should be prevented.
  - **Scalability:** The system should support multiple users simultaneously without performance degradation.
  - **Reliability:** The system should function consistently without crashes or data loss.
  - **Compatibility:** The application should run smoothly on different mobile devices and operating systems.
  - **Maintainability:** The system should be easy to update and extend with new features or experiments.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the proposed system includes school students (Grade 6–11), particularly in regions where access to physical laboratory facilities is limited. Secondary markets include schools, tuition centers, and educational institutions seeking digital learning solutions. The key customer persona is a student who struggles with understanding physics concepts due to lack of practical exposure, as well as teachers who require interactive tools to improve student engagement and learning outcomes. Additionally, education authorities and private learning platforms can adopt the system for wider implementation.

Value Proposition

The proposed system offers a unique value by combining game-based learning, real-time measurement generation, automated physics calculations, and virtual experiment evaluation in a single mobile application. Unlike existing solutions that focus on only one aspect (simulation, games, or labs), this system provides a comprehensive and interactive learning experience**.** It improves conceptual understanding, increases student engagement, and provides accessible practical learning without requiring physical laboratory infrastructure.

Revenue Model

The revenue model for this system can be based on a freemium educational platform approach. Basic features such as access to selected physics games and learning modules can be offered for free to encourage adoption by schools and students. Advanced features, including additional experiments, detailed analytics, and extended learning modules, can be provided through subscription-based plans for schools or institutions. Schools may subscribe to a yearly or monthly plan to access the full platform for their students. Additional revenue opportunities may also include institutional licensing, partnerships with educational organizations, and integration with digital learning programs**.**

Cost Estimation

The main costs associated with developing and maintaining the system include software development, system maintenance, cloud services, and content creation**.** Development costs involve building the mobile application using Flutter, developing backend services using Python, integrating Firebase for cloud database management, and creating interactive games and simulations using Unity 2D. Additional costs may include cloud hosting services, security implementation, testing, and continuous updates. Operational costs may also include technical support, server maintenance, and future development for adding new experiments or learning features.

Pricing Strategy

The pricing strategy should focus on making the platform affordable and accessible for educational institutions, particularly schools in developing regions. **A** tiered pricing model can be used where individual users have access to basic features for free, while schools can purchase institutional packages that allow multiple students and teachers to use the system. Pricing may include monthly or yearly subscription plans for schools, with discounts for larger institutions or government educational programs. This strategy ensures that the system remains accessible to a wide range of users while still supporting sustainable platform development.

Competitive Advantage

The proposed system has several competitive advantages compared to existing educational platforms. First, it integrates physics-based games, virtual experiments, real-time measurement generation, and step-by-step calculation explanations within a single learning environment**,** which many current platforms do not provide. Second, the mobile-based design makes the platform accessible and easy to use for students in rural areas where laboratory facilities may be limited. Third, the system focuses on transforming student interaction into measurable physics learning outcomes, allowing students to understand concepts through observation, experimentation, and calculation rather than memorization.

Intellectual Property Considerations

The developed system, including software design, algorithms, and user interface, can be protected under copyright laws. Any unique methods related to measurement extraction, calculation generation, or evaluation processes may be considered for intellectual property protection. Proper licensing will be ensured for third-party tools and libraries such as OpenCV and Firebase. This helps secure ownership and supports future commercialization opportunities.

7. BUDGET AND JUSTIFICATION

- **Hardware costs**
  The proposed research does not require any specialized hardware devices, as the system is developed as a mobile-based software application. Development and testing activities can be carried out using existing resources such as personal laptops and smartphones owned by the research team. Since the system focuses on virtual simulations and digital experiments rather than physical laboratory equipment, additional hardware purchases are not necessary. Therefore, the estimated hardware cost for this project is minimal or zero.

- **Software licenses**
  The project utilizes mainly open-source and free development tools. Flutter is used for mobile application development, Python is used for backend processing, and Firebase provides a free-tier cloud service suitable for small-scale applications. Unity 2D also offers a free personal license for educational purposes. OpenCV and YOLOv8 are open-source technologies used for computer vision and object detection. As a result, no significant cost is required for software licenses, making the project highly cost-effective.

- **Cloud services**
  The system uses Firebase as the cloud platform for data storage, authentication, and real-time database management. Firebase offers a free-tier plan that is sufficient for development, testing, and demonstration purposes in this research. This allows the system to operate efficiently without incurring additional cloud service costs during the research phase. Therefore, cloud service expenses are expected to be minimal or zero.

- **Human effort cost**
  The development, design, and implementation of the system are carried out by the research team as part of their academic research work. Since this project is conducted within an educational context, human effort is not calculated as a financial cost. Instead, it is considered as a contribution of time, knowledge, and skills by the students and supervisors involved in the project.

- **Justification for each expense**
  The proposed system is designed to be highly cost-effective by utilizing existing devices and open-source technologies. The absence of hardware requirements and the use of free development tools significantly reduce the overall budget. Firebase's free cloud services further minimize operational costs. This approach demonstrates that the system can be developed and implemented with minimal financial investment, making it practical and scalable for educational institutions, especially in resource-limited environments. Additionally, the use of widely available technologies ensures that the system can be easily maintained, extended, and adopted in the future without requiring significant financial resources.

Table 1 Budget and Budget Justification

Budget Category	**Item/Resource**	**Estimated Cost (LKR)**	**Justification**
Hardware Costs	Existing laptops and smartphones	0	Development and testing will use personal devices owned by the research team. No additional hardware is required.
Software Licenses	Flutter, Python, Unity 2D, OpenCV, YOLOv8	0	All tools are open-source or free for educational use
Cloud Services	Firebase Free Tier	0	Firebase provides free database, authentication, and cloud services sufficient for development and testing.
Development Tools	Visual Studio Code	0	These development environments are freely available for software development
Human Effort Cost	Student research team work		The development and implementation are part of the academic research work of the students.
Total Estimated Budget		0	

8. WORK BREAKDOWN STRUCTURE (WBS)

Table 2 Work Breakdown Structure

WBS ID	Task Name	Member	Description	Duration
1.0	Project Initiation	All	Identify research problems, define project scope, and obtain supervisor approval	Month 1
1.1	Literature Review	All	Review research papers related to virtual labs, mobile learning, learning analytics, adaptive feedback, RAG-based chatbots, gamification, and personalized learning systems	Month 1 – Month 2
1.2	Research Proposal Preparation	All	Prepare research proposal including introduction, research gaps, objectives, and methodology for all four modules	Month 2
2.0	System Requirement Analysis	All	Identify functional and non-functional requirements of the complete mobile virtual physics laboratory system	Month 3
2.1	System Architecture Design	All	Design overall system architecture including Flutter mobile application, Python backend services, Firebase, and database structure	Month 3
2.2	Database Design	All	Design Cloud Firestore data structure for user profiles, quiz scores, experiment data, analytics records, chatbot data, game records, learning paths, and curriculum content	Month 3
3.0	User Authentication and Profile Module	Wijerathna C D W	Implement student registration, login, Google Sign-In, guest access, and student profile management using Firebase Authentication and Cloud Firestore	Month 4

3.1	Intelligent Quiz Evaluation System	Wijerathna C D W	Develop intelligently generated sub-lesson quizzes with MCQs and structured questions aligned to Grade 6–11 curriculum, with shuffled retake support	Month 5 – Month 6
3.2	Experiment Performance Evaluation	Wijerathna C D W	Implement system to evaluate practical experiment steps, record performance outcomes, and identify poorly performed experiment areas per sub-lesson	Month 6 – Month 7
3.3	Learning Analytics and Weakness Detection	Wijerathna C D W	Develop algorithms to analyze sub-lesson quiz and experiment results and detect weak theory and practical areas	Month 7 – Month 8
3.4	Adaptive Feedback Generation	Wijerathna C D W	Implement rule-based feedback mechanism to generate personalized feedback messages and display improvement suggestions on the student home page	Month 8
3.5	Personalized Learning Path Assignment	Wijerathna C D W	Develop learning path assignment system that detects weak students and assigns targeted resources including explanations, guided examples, experiment procedures, and video materials; enable reattempt after completion; assign additional external resources for persistent weakness	Month 8 – Month 9
3.6	Dashboard and Visualization Development	Wijerathna C D W	Develop student dashboard with sub-lesson performance indicators, quiz and experiment scores, weak area summaries, feedback messages, and learning path access	Month 9 – Month 10
4.0	Grade-Tagged Physics Knowledge Base	Sachintha Nirmal S.M	Build a grade-tagged physics knowledge base containing curriculum-aligned physics content organized by grade level,	Month 4 – Month 5

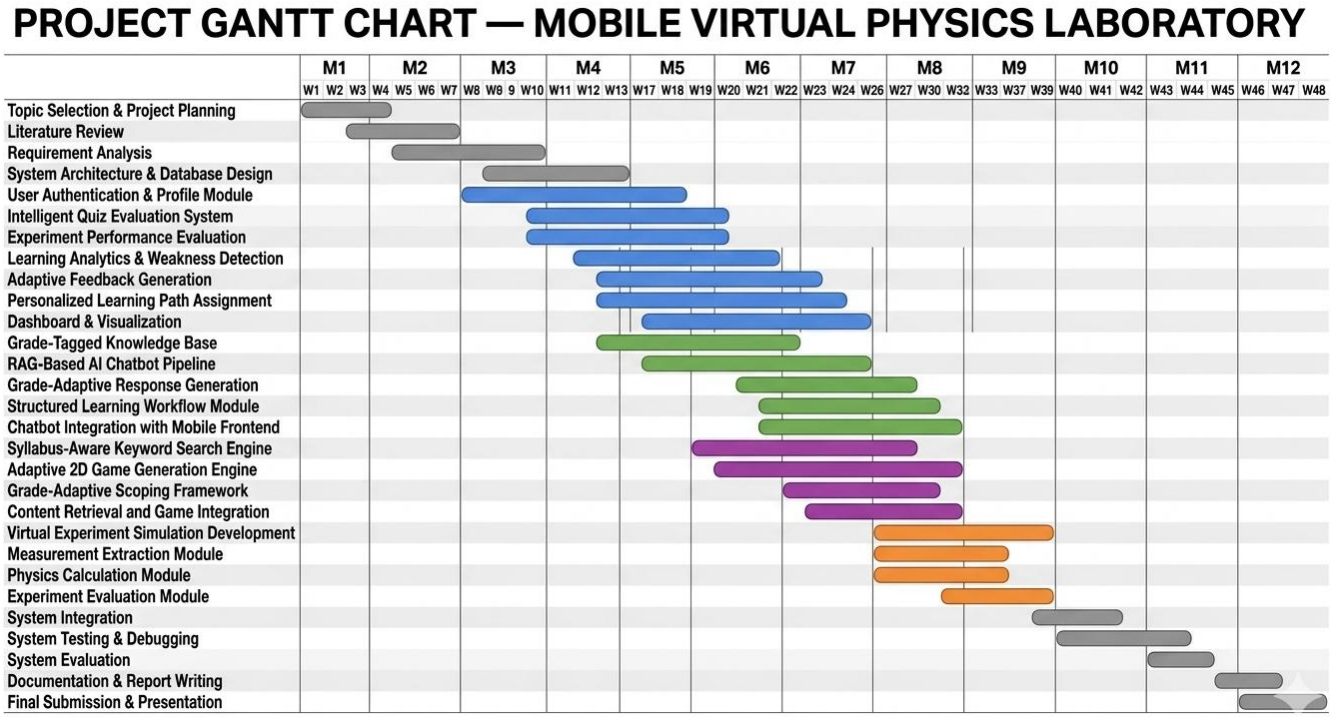
			lesson, topic, and activity for RAG retrieval	
4.1	RAG-Based AI Chatbot Pipeline	Sachintha Nirmal S.M	Implement the Retrieval-Augmented Generation pipeline using LLM to retrieve relevant syllabus content and generate physics explanations for student queries	Month 5 – Month 6
4.2	Grade-Adaptive Response Generation	Sachintha Nirmal S.M	Develop grade-adaptive response generation that tailors chatbot answers based on the student's grade level and lesson context	Month 6
4.3	Structured Learning Workflow Module	Sachintha Nirmal S.M	Design and implement the structured learning workflow including hierarchical navigation for grades, lessons, topics, experiments, and activities	Month 4 – Month 5
4.4	Chatbot Integration with Mobile Frontend	Sachintha Nirmal S.M	Integrate the AI chatbot interface and structured learning workflow screens into the Flutter mobile application	Month 7
5.0	Syllabus-Aware Keyword Search Engine	Arthanayake C T H	Implement keyword-based search engine that matches student-entered physics keywords with Sri Lankan National curriculum syllabus materials	Month 4 – Month 6
5.1	Adaptive 2D Game Generation Engine	Arthanayake C T H	Create curriculum-aligned 2D educational games using Unity that adapt game difficulty and content based on student grade level and lesson topic	Month 5 – Month 7
5.2	Grade-Adaptive Scoping Framework	Arthanayake C T H	Design scoping framework that maps digital learning activities and game difficulty to structured lesson-level objectives for Grades 6–11	Month 6 – Month 7

5.3	Content Retrieval and Game Integration	Arthanayake C T H	Integrate keyword-based retrieval system and game generation module with Firebase Firestore for NIE-aligned curriculum data storage and retrieval	Month 7 – Month 8
6.0	Virtual Experiment Simulation Development	Vidurangi H I D G	Develop interactive physics virtual experiments using Unity 2D where students can interact with virtual objects and perform guided experiments	Month 4 – Month 7
6.1	Measurement Extraction Module	Vidurangi H I D G	Implement real-time measurement extraction system using OpenCV and YOLOv8 to capture physics measurements such as distance, time, and velocity from student interactions	Month 5 – Month 6
6.2	Physics Calculation Module	Vidurangi H I D G	Apply physics formulas to extracted measurements to generate step-by-step calculation results and concept explanations for students	Month 6 – Month 7
6.3	Experiment Performance Evaluation Module	Vidurangi H I D G	Implement scoring system that evaluates student experiment performance based on safety tool usage, apparatus selection, procedure steps, and final result accuracy	Month 7 – Month 8
7.0	System Integration	All	Integrate Flutter mobile application, Python backend analytics services, Firebase Authentication, Cloud Firestore, chatbot, experiment module, gamified learning, and analytics module into a complete unified system	Month 10 – Month 11
8.0	System Testing and Debugging	All	Conduct functional and performance testing of all modules including authentication, quiz evaluation, experiment evaluation, weakness detection, feedback	Month 10 – Month 11

			generation, learning path assignment, chatbot accuracy, RAG retrieval, experiment simulations, and game functionality	
9.0	System Evaluation	All	Evaluate usability and effectiveness of the complete system including mastery level prediction accuracy, chatbot performance, RAG retrieval accuracy, experiment evaluation accuracy, search relevance, game engagement, and learning path effectiveness	Month 11
10.0	Documentation and Final Report	All	Prepare final research documentation including results, discussion, and complete system description for all four modules	Month 11 – Month 12
11.0	Final Presentation Preparation	All	Prepare presentation slides, demonstration materials, and final system submission	Month 12

9. GANTT CHART

Table 3 Gantt Chart



10. References List

[1] H. J. Banda and J. Nzabahimana, "When Learning About the Real World is Better Done Virtually:," *The Impact of Interactive Simulations on Students' Conceptual Understanding in Physics Education,* vol. 12, no. 12, pp. 1 - 15, 2022.

[2] C. E. Wieman, W. K. Adams and K. K. Perkins, "PhET: Simulations That Enhance Learning," *PhET: Simulations That Enhance Learning,* vol. 322, no. 682, pp. 1 - 5, 2008.

[3] N. D. Finekelstein, W. K. Adams, C. J. Keller, P. B. Kohl, K. K. Perkins, N. S. Podolefsky, S. Reid and R. LeMatser, "When learning about the real world is better done virtually: A study of substituting computer," *The Physics Teacher,* vol. 48, no. 4, pp. 1 - 8, 2005.

[4] A. N. Foster, M. J. Koeher and P. Mishra, "Game-Based Learning of Physics Content: The Effectiveness of a Physics Game for Learning Basic Physics Concepts," *Physics Game for Learning Basic Physics Concepts,* Vols. -, no. -, pp. 1 - 7, 2006.

[5] R. C. Sanchez, A. M. Leon, J. M. Rodriguez-Ferrer, J. Serna and P. L. Burgues, "Game-Based Learning and Gamification in Physical Education:A Systematic Review," *A Systematic Review,* vol. 13, no. 183, pp. 1 - 12, 2023.

[6] O. D. Pranata, "Physics education technology (PhET) as a game-based learning tool:A quasi-experimental study," *A quasi-experimental study,* vol. 9, no. 4, pp. 1 - 11, 2024.

[7] A. U. Dy, J. C. Lagura and G. R. Baluyos, "Using PhET Interactive Simulations to Improve the Learners'," *Science,* vol. 4, no. 4, pp. 1 - 11, 2024.

[8] M. N. Mohd Najib, A. Yaacob and R. Md-Ali, "Exploring the Effectiveness of Interactive Simulation as Blended Learning Approach in Secondary School Physics," *Blended Learning Approach in Secondary School Physics,* vol. 82, no. 1, pp. 1 - 9, 2022.

[9] A. Filanovich, "Virtual Laboratories in Physics Education," *International Journal of Modern Education and Computer Science,* vol. 14, no. 3, pp. 1 - 8, 2022.

[10] C. Dichev and D. Dicheva, "Gamifying Education:What is known, whta is believed and what remains uncertain:a critical review," *International Journal of Educational Technology in Higher Education,* vol. 14, no. 9, pp. 1 -36, 2017.

11.APPENDICES

APPENDIX 1 Survey instruments

This research does not include survey-based data collection methods. Instead, the system evaluation is conducted through user interaction testing and performance analysis. Therefore, no survey instruments are included in this study.

APPENDIX 2 Detailed diagrams

This section includes detailed diagrams related to the system design and methodology. These diagrams provide a visual representation of the research workflow and system structure. Key diagrams include:

- Research Methodology Diagram
- Research Methodology Process Diagram
- High-Level System Architecture Diagram

These diagrams illustrate the flow of activities, system components, and interactions between modules such as the mobile application, game engine, measurement module, calculation engine, and cloud database.

APPENDIX 3 Preliminary UI mockups

Preliminary user interface (UI) mockups are included to demonstrate the design of the mobile application. These mockups present the layout and structure of key screens such as:

- Home screen (navigation to games and experiments)
- Physics game interface (interaction and measurement display)
- Calculation screen (step-by-step explanation of formulas)
- Virtual laboratory interface (experiment setup and tools)
- Result and feedback screen

These mockups help visualize how users will interact with the system and ensure a user-friendly design suitable for school students.

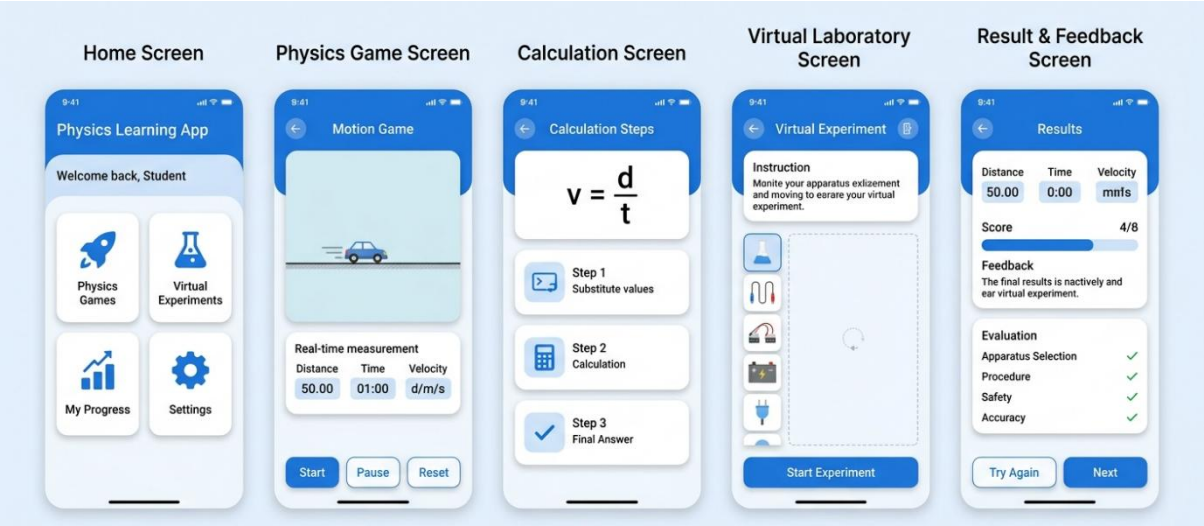


Figure 04 Preliminary UI mockups

APPENDIX 4 Risk analysis

Table 4 Risk Analysis

Risk Type	Description	Mitigation Strategy
Technical Risk	Challenges in accurate measurement extraction using computer vision	Use optimized models(YOLOv8) and test with multiple scenarios
Performance Risk	Possible delays in real-time processing	Optimize algorithms and use efficient backend processing
User Adoption Risk	Students or teachers may be unfamiliar with the system	Provide simple UI design and user guidance
Data Security Risk	Risk of unauthorized access to user data	Use secure Firebase authentication and data protection mechanisms
System Reliability Risk	Application crashes or unexpected bugs	Perform thorough testing and continuous updates

APPENDIX 5 Ethical documents

Ethical considerations are maintained throughout the research and system implementation. The following principles are followed:

- User data is collected only for educational and research purposes.
- No sensitive personal information is stored in the system.
- User privacy and confidentiality are ensured through secure data handling.
- Participation in system testing is voluntary, and consent is obtained where required.
- The system is designed to support fair and inclusive access to education without bias

Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students

Project ID: R26-IT-150

Project Proposal Report

ARTHANAYAKE C T H

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

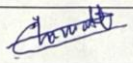
Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Arthanayake C T H	IT22251800	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Isuri Dekewatte	18/03/2026	

Scanned with CamScanner

ABSTRACT

Rural school students in Sri Lanka face a persistent gap between theoretical physics knowledge and the ability to explore concepts interactively. The majority of government schools in rural and semi-urban areas lack laboratory infrastructure, curriculum-aligned digital reference materials, and adaptive learning experiences that match the student's specific grade level. Existing virtual laboratory and educational technology platforms either provide static simulations disconnected from the Sri Lankan NIE curriculum, generic search tools not scoped to grade level, or entertainment-focused games that do not produce real physics measurements or enforce conceptual mastery progression.

This research proposes a two-part intelligent learning module within the PhysiV-Lab mobile virtual physics laboratory system, targeting Grade 6-11 students. The first sub-system is an Intelligent Keyword-Based Content Retrieval engine that implements a two-tier architecture: a BM25 baseline for fast keyword matching and a fine-tuned Sentence-BERT semantic search model trained on NIE physics content, both filtered by the student's grade level. The second sub-system is a Curriculum-Aligned 2D Game Generation framework that uses a template-based Unity architecture driven by Firebase configuration data to deliver mastery-based progressive physics gameplay scoped to the NIE syllabus at each grade level, producing real-time physics measurements and formula explanations from student interactions.

Technologies used include Python FastAPI for the backend, rank_bm25 and sentence-transformers for retrieval, Google Colab for model training, Flutter with flutter_unity_widget for mobile integration, Unity 2D for game scenes, and Firebase Firestore for content storage and game configuration. The NIE physics content dataset is constructed manually by the researcher from official NIE textbooks. The SciQ dataset is used for initial domain adaptation of the SBERT model.

Expected outcomes include a deployed grade-adaptive semantic retrieval system validated against a held-out NIE query set, and a functional curriculum-aligned game module with three progressive difficulty stages validated for measurement accuracy, calculation correctness, and grade-difficulty alignment. Together these sub-systems form a Grade-Responsive Syllabus-Aware Intelligent Search and Adaptive 2D Gameplay-Level Scoping Framework unique to Sri Lankan school physics education. The component supports SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities).

Keywords: Semantic Content Retrieval, Sentence-BERT, BM25, Curriculum-Aligned Game Generation, Mastery-Based Progression, Mobile Physics Education, NIE Syllabus

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor, Mr. Indunil Daluwatte, for his continuous guidance, encouragement, and constructive feedback throughout the development of this research proposal. His insights into educational technology and mobile system design have greatly shaped the direction of this work.

I extend my appreciation to the academic staff of the Department of Information Technology at the Sri Lanka Institute of Information Technology for providing the academic foundation and resources necessary to conduct this research. I am grateful to my fellow team members for their collaboration during system design discussions.

Finally, I wish to express my deepest appreciation to my family and friends for their unwavering support and encouragement throughout my academic journey.

TABLE OF CONTENT

Contents

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT ..... **Error! Bookmark not defined.**

TABLE OF CONTENT 5

LIST OF TABLES 7

LIST OF FIGURES 7

LIST OF ABBREVIATIONS 7

LIST OF APPENDICES 7

1. INTRODUCTION 8

1.1 Background and Context 8

1.2 Literature Review 9

1.3 Research Gap 10

2. OBJECTIVE 12

2.1 Main Objective 12

2.2 Specific Objectives 12

3. METHODOLOGY 13

3.1 Research Approach 13

3.2 Sub-System 1: Intelligent Keyword-Based Content Retrieval 13

3.3 Sub-System 2: Curriculum-Aligned 2D Game Generation 14

3.4 Evaluation Strategy 16

3.5 Technology Stack 16

3.6 Ethical Considerations 17

4. HIGH-LEVEL SYSTEM ARCHITECTURE 18

4.1 Project Structure and Component Scope 18

4.2 Component Position in Lesson Workflow 18

4.3 System Architecture Diagram 18

4.4 Data Flow 19

4.5 Scalability and Extensibility 19

5. USER REQUIREMENTS 20

5.1 Stakeholder Identification 20

5.2 Functional Requirements 20

5.3 Non-Functional Requirements 21
6. COMMERCIALIZATION PLAN 21
6.1 Target Market and Customer Persona 21
6.2 Value Proposition 22
6.3 Revenue Model 22
6.4 Competitive Advantage 22
6.5 Intellectual Property Considerations 22
7. BUDGET AND JUSTIFICATION 23
8. WORK BREAKDOWN STRUCTURE (WBS) 24
9. GANTT CHART 26
10. REFERENCES LIST 27
11. APPENDICES 28

LIST OF TABLES

Table 1:Research Gap Analysis 11
Table 2:NIE Physics Content Dataset Structure 13
Table 3:Game Template Descriptions 15
Table 4: Stage Design Summary 15
Table 5:Difficulty Scaling Parameters by Grade 15
Table 6:Planned Evaluation Metrics and Intended Benchmarks 16
Table 7:Stakholder identification 20
Table 8:Functional Requirements 21
Table 9: Non-Functional Requirements 21
Table 10:Budget summery 23
Table 11:Work Breakdown Structure 25
Table 12:NIE physics content dataset ..... **Error! Bookmark not defined.**
Table 13:Linear Motion Lab game template ..... **Error! Bookmark not defined.**
Table 14:Risk Analysis 30

LIST OF FIGURES

Figure 1:System Architecture Diagram 18
Figure 2: System flow. 19
Figure 3: Grant chart 26

LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
BM25	Best Match 25 (Information Retrieval Algorithm)
MRR	Mean Reciprocal Rank
NIE	National Institute of Education (Sri Lanka)
NLP	Natural Language Processing
SBERT	Sentence Bidirectional Encoder Representations from Transformers
SDG	Sustainable Development Goal
SUS	System Usability Scale
TF-IDF	Term Frequency – Inverse Document Frequency
UI	User Interface

LIST OF APPENDICES

Preliminary Feasibility Study 29

NIE Physics Content Dataset Sample Entries 29

1. INTRODUCTION

1.1 Background and Context

Real-World Problem

Physics education in rural Sri Lanka is significantly constrained by a lack of laboratory infrastructure, limited curriculum-aligned digital learning resources, and an absence of adaptive interactive experiences that match the specific grade level of the learner [1], [3]. Students studying physics under the Sri Lankan National Institute of Education (NIE) curriculum for Grades 6 to 11 frequently encounter unfamiliar concepts during self-study with no accessible, curriculum-accurate reference they can query from a mobile device. At the same time, while virtual laboratory platforms and educational games exist globally, none are aligned to the Sri Lankan NIE syllabus, adapted to specific grade levels, or capable of producing real physics measurements from student interactions rather than purely entertainment-driven outcomes.

Two specific gaps define the problem addressed by this component. First, when a student encounters an unfamiliar term such as velocity, pressure, or Ohm's law during a lesson, they have no immediate lightweight search tool scoped to their grade level and curriculum. General internet searches return content at inappropriate difficulty levels and are not aligned to the NIE syllabus. Second, while gamified learning has demonstrated strong potential for improving motivation and retention in science education [1], existing educational games do not convert student gameplay actions into real-time physics measurements, do not enforce mastery-based stage progression, and are not driven by NIE curriculum parameters.

Who is Affected and Why It Matters

The students most significantly affected are the approximately 1.2 million learners enrolled in Grades 6 to 11 in government schools across Sri Lanka, a disproportionate share of whom study in rural schools without laboratory access [3]. Physics consistently registers among the weakest performance areas in national examinations, and the inability to access curriculum-aligned reference materials or engage with concept-reinforcing interactive activities during self-study is a documented contributor to this outcome [3], [5]. Providing lightweight, grade-filtered retrieval and curriculum-aligned gameplay directly addresses both dimensions of this problem within a single mobile application.

Statistical Evidence

Kalogiannakis et al. [1] demonstrate through systematic review of 96 studies that gamification in science education significantly improves both motivation and conceptual retention, particularly where gameplay is directly connected to measurable learning objectives. Elmoazen et al. [5] show that virtual laboratory learning analytics systems consistently improve learning outcomes where student interactions are linked to real physics measurements rather than passive simulation observation. Vijayatheepan [3] documents that virtual laboratory access is unavailable in the majority of rural Sri Lankan government schools, while also noting that teachers express strong demand for curriculum-aligned digital tools that can operate on low-cost mobile devices.

Analysis of Existing Solutions and Their Limitations

PhET Interactive Simulations and Labster provide high-quality virtual experiment environments but are not aligned to the Sri Lankan NIE curriculum, do not offer grade-filtered keyword search, and are not optimised for mobile devices with limited connectivity [5]. General educational search tools such as Khan Academy and Google Search return results across multiple curricula and difficulty levels without grade-level scoping. Educational game platforms such as Kahoot and Quizziz offer gamified quiz formats but do not produce physics measurements from student interactions, do not enforce mastery-based stage progression, and are not driven by curriculum configuration data [1]. Retrieval systems proposed in academic research such as Gil et al. [2] demonstrate keyword-based educational content retrieval but are designed as standalone systems rather than integrated lesson components, and do not support semantic search or NIE curriculum alignment.

SDG Alignment

This component supports SDG 4 (Quality Education) by providing curriculum-aligned retrieval and interactive physics learning for underserved rural students. It supports SDG 10 (Reduced Inequalities) by delivering equal access to intelligent educational tools regardless of geographic location. The research is aligned with the Technology Integration and Management (TIM) research cluster and the Information Technology (IT) specialization.

1.2 Literature Review

A review of twelve recent scholarly sources was conducted to examine the current state of research in educational content retrieval, sentence embedding models, gamification in science education, virtual laboratory learning, and curriculum-aligned game design.

Kalogiannakis et al. [1] provide the most comprehensive systematic review of gamification in science education to date, analysing 96 empirical studies and concluding that gamification significantly improves learner motivation, engagement, and conceptual retention in STEM subjects. Critically, the review identifies that the most effective interventions connect game mechanics directly to measurable scientific outcomes rather than using entertainment as a surface overlay on passive content. This finding directly motivates the measurement-driven game design in this component, where student gameplay actions produce real physics calculations rather than arbitrary scores.

Gil et al. [2] propose a smart retrieval system for digital educational content using machine learning, demonstrating that keyword-based retrieval with ranking algorithms can effectively match student queries to relevant educational materials. Their architecture achieves strong precision at retrieval but does not support semantic understanding of paraphrased queries, does not apply grade-level filtering, and is not designed for integration within a mobile lesson workflow. This limitation motivates the two-tier architecture in this component, where BM25 handles exact matches and fine-tuned SBERT handles semantic paraphrases.

Reimers and Gurevych [4] introduced Sentence-BERT (SBERT), demonstrating that siamese BERT networks produce semantically meaningful sentence embeddings suitable for semantic textual similarity and information retrieval tasks. The all-MiniLM-L6-v2 variant used in this component is an 80MB distilled model validated for fast and accurate semantic search tasks, making it suitable for deployment on a Python backend without GPU requirements. Fine-tuning SBERT on domain-specific

content has been shown to improve retrieval accuracy substantially over the base model in specialised domains [4].

Elmoazen et al. [5] conduct a systematic review of learning analytics in virtual laboratory environments and conclude that the most educationally effective virtual lab systems are those that capture real physics measurements from student interactions and use them to drive formative feedback. Systems that log only activity events without producing measurement-linked outcomes provide insufficient data for learning analytics and do not support meaningful assessment. This finding directly justifies the measurement-driven game design in Sub-System 2, where every student interaction produces a physics calculation that is stored and passed to the analytics module.

Robertson and Zaragoza [10] formally established BM25 as the standard probabilistic ranking algorithm for information retrieval, demonstrating consistent superiority over TF-IDF and providing the theoretical basis for its use as the Tier 1 baseline in this component. The SciQ dataset [8], comprising 13,679 crowd-sourced science questions and answers from school-level subjects, provides the initial domain adaptation corpus for SBERT fine-tuning before NIE-specific fine-tuning is applied. Hugging Face's sentence-transformers library [4] provides the training framework and pre-trained weights.

In the domain of game-based learning, research on mastery-based progression demonstrates that enforcing minimum performance thresholds before unlocking subsequent stages produces significantly stronger conceptual transfer than free-progression designs, particularly in STEM subjects [1], [6]. Template-based game architecture in educational technology has been validated as a scalable approach to curriculum-aligned game delivery, where a small number of reusable game scenes are driven by configuration data rather than requiring new development for each topic or grade level [6]. The Unity game engine combined with the flutter_unity_widget library has been demonstrated as a viable architecture for embedding interactive 2D games within Flutter mobile applications [7].

1.3 Research Gap

Despite advances in educational content retrieval, semantic search models, and game-based science learning, a clear gap exists in the integration of these capabilities within a single curriculum-aligned mobile lesson component for Sri Lankan school physics.

No existing system provides grade-filtered semantic content retrieval aligned to the Sri Lankan NIE physics syllabus. Existing retrieval systems either operate at a general difficulty level across multiple curricula, or use keyword matching without semantic understanding. The combination of a BM25 baseline and a NIE-fine-tuned SBERT semantic model with grade-level filtering has not been proposed or evaluated for school-level physics retrieval in a Sri Lankan or comparable developing-country context.

In the domain of game-based physics learning, no existing platform delivers curriculum-aligned 2D physics games driven by Firebase configuration data that produce real-time physics measurements from student interactions, enforce mastery-based stage progression across three difficulty levels, and scale difficulty parameters automatically to the NIE grade level without requiring Unity code changes. The combination of template-based game architecture, Firebase-driven configuration, measurement-linked student interactions, and NIE curriculum alignment represents a contribution not present in any reviewed system.

Dimension	**Gap in Existing Literature**	**This Study Aims To Investigate**
Content retrieval scope	Retrieval systems are not scoped to NIE curriculum or grade level	Whether grade-filtered BM25 + SBERT retrieval over NIE content

		produces accurate curriculum-aligned results
Semantic understanding	BM25-only systems fail on paraphrased or indirect queries	Whether SBERT fine-tuned on NIE physics content improves retrieval over BM25 baseline for indirect queries
Game measurement linkage	Educational games produce scores without real physics measurements	Whether template-based games producing real-time physics calculations improve conceptual understanding over score-only games
Mastery progression	Existing game platforms allow free progression without threshold enforcement	Whether mastery-based stage gating improves physics concept retention compared to free-progression designs
Grade difficulty scaling	Game difficulty is static or manually adjusted per grade	Whether Firebase-driven dynamic difficulty scaling to NIE grade parameters is feasible without Unity code modification
Curriculum artefact	No structured digital NIE physics content dataset exists for Grades 6-11	Whether a manually curated NIE query-document dataset enables effective SBERT fine-tuning for curriculum-specific retrieval

Table 1:Research Gap Analysis

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate a two-part intelligent learning support module comprising a grade-adaptive two-tier semantic content retrieval system aligned to the Sri Lankan NIE physics curriculum and a curriculum-aligned 2D game generation framework that dynamically scopes gameplay difficulty, physics measurements, and formula explanations to the student's NIE grade level, within a mobile virtual physics laboratory application for Grade 6-11 students in rural Sri Lanka.

2.2 Specific Objectives

1. To construct a structured NIE-aligned physics content dataset comprising 200 to 400 query-document pairs across Grades 6 to 11, to serve as the retrieval corpus for BM25 indexing and SBERT fine-tuning.
2. To implement and evaluate a Tier 1 BM25 keyword retrieval system over the NIE content dataset with grade-level filtering, and establish its performance as a baseline using Precision@K, Recall@K, and MRR metrics.
3. To fine-tune a Sentence-BERT model on the NIE physics content dataset and evaluate whether semantic retrieval improves over the BM25 baseline for indirect and paraphrased student queries.
4. To design and implement a template-based Unity 2D game architecture driven by Firebase configuration data that delivers curriculum-aligned physics gameplay across a minimum of three game templates covering core NIE physics topics.
5. To implement and evaluate a three-stage mastery-based progression system within each game template, enforcing grade-appropriate pass thresholds and producing real-time physics measurements and step-by-step formula explanations from student interactions.
6. To validate the complete two-sub-system component through functional testing, retrieval quality evaluation, game measurement accuracy testing, and usability evaluation with student participants where feasible.

3. METHODOLOGY

3.1 Research Approach

This study adopts a Design Science Research methodology, which is well established in Information Systems research for designing, implementing, and evaluating software artefacts that address identified practical problems [9]. The study will proceed through four phases: problem identification and requirements analysis, artefact construction and system design, prototype implementation, and experimental evaluation. Evaluation will combine functional testing, retrieval quality benchmarking on a held-out query set, game measurement accuracy testing, and usability evaluation with student participants where feasible.

3.2 Sub-System 1: Intelligent Keyword-Based Content Retrieval

NIE Physics Content Dataset Construction

The primary dataset is constructed manually by the researcher from official Sri Lankan NIE Grade 6-11 physics textbooks, freely available at nie.lk. Each document entry covers one physics concept at one grade level and follows a structured schema containing a concept identifier, grade level, topic title, plain-language explanation, formula where applicable, worked example, and a set of keyword tags. For each document, two to four representative query strings are authored to reflect the natural language a student might use when searching for that concept. The target dataset size is 200 to 400 query-document pairs across all six grades and core NIE physics topics. This artefact constitutes an original research contribution as no equivalent structured digital dataset for Sri Lankan secondary physics currently exists in the public domain.

Field	Description
concept_id	Unique identifier for the physics concept
grade	Integer grade level (6 to 11)
topic	NIE lesson topic title
explanation	Plain-language explanation appropriate for the grade level
formula	Physics formula where applicable, with variable definitions
example	One worked numerical example at the appropriate grade level
keywords	Array of keyword tags for BM25 indexing
query_strings	Array of 2 to 4 representative natural language queries

Table 2:NIE Physics Content Dataset Structure

The SciQ dataset [8], comprising 13,679 school-level science questions and answers, is used as a supporting corpus for initial domain adaptation of the SBERT model before NIE-specific fine-tuning is applied. No private or sensitive data is collected or processed.

Tier 1: BM25 Keyword Retrieval

The baseline retrieval approach uses BM25 via the rank_bm25 Python library. The NIE content documents are indexed at startup of the FastAPI backend. When a student submits a query, the backend first applies a grade-level filter to restrict the candidate document set to entries matching the student's grade, then applies BM25 ranking over the filtered set. The top-ranked document is returned if its BM25 score exceeds a defined confidence threshold. This approach requires no model training, runs entirely on CPU, produces reliable results for exact and near-exact keyword queries, and serves as the performance baseline against which SBERT is compared.

Tier 2: Fine-Tuned Sentence-BERT Semantic Retrieval

The enhanced retrieval approach fine-tunes the all-MiniLM-L6-v2 Sentence-BERT model from HuggingFace using the sentence-transformers library. Fine-tuning proceeds in two stages. In Stage 1, the base model is fine-tuned on the SciQ dataset using a MultipleNegativesRankingLoss objective to adapt it to the school science domain. In Stage 2, the domain-adapted model is further fine-tuned on the NIE query-document pairs to specialise it for Sri Lankan curriculum content. Training is conducted on Google Colab free tier, which provides sufficient compute for a dataset of this size without GPU purchase. The trained model weights are exported and deployed on the Python FastAPI backend. At retrieval time, the student query and all grade-filtered candidate documents are encoded using the model, and cosine similarity ranking is applied to return the most semantically relevant result.

Retrieval Decision Logic

At query time, the system applies Tier 1 BM25 first. If the top BM25 score exceeds a defined threshold, the result is returned directly without invoking SBERT, minimising response latency. If BM25 confidence is below threshold, indicating a likely paraphrased or indirect query, Tier 2 SBERT semantic ranking is applied over the grade-filtered candidate set and the cosine similarity top result is returned. This cascaded approach ensures fast responses for common keyword queries while providing semantic fallback for indirect phrasing.

3.3 Sub-System 2: Curriculum-Aligned 2D Game Generation

Template-Based Architecture

Rather than building a separate Unity game for each topic and grade level, the system uses a template-based architecture comprising a small number of reusable Unity 2D game scenes whose content, difficulty parameters, and physics calculations are driven entirely by Firebase Firestore configuration data loaded at runtime. This design means that adding a new topic or adjusting difficulty for a grade requires only a new Firestore document, with no changes to Unity code. Three initial game templates are planned, covering the most frequently assessed physics topics in the NIE curriculum.

Template Name	Physics Concept	Student Interaction
Linear Motion Lab	Speed, Velocity, Distance, Time	Student sets distance or time values; virtual vehicle moves; system calculates and validates speed

Force and Mass Arena	Newton's Second Law, Force, Acceleration	Student applies force to objects of different masses; system measures resulting acceleration and validates F = ma
Pressure Chamber	Fluid Pressure, Area, Force	Student adjusts force or area values on a piston; system calculates resulting pressure and validates P = F/A

Table 3:Game Template Descriptions

Three-Stage Mastery Progression

Every game template implements three progressive stages. Students must achieve the minimum score threshold for each stage before the next stage is unlocked, enforcing mastery-based progression. This design is motivated by research demonstrating that threshold-gated progression produces stronger conceptual transfer than free-progression designs in STEM game-based learning [1].

Stage	Name	Design	Pass Threshold
Stage 1	**Tutorial (Guided)**	All values provided. Formula visible on screen. Student observes real-time physics calculation. No time limit. Hint button available.	No threshold — observational
Stage 2	**Core (Interactive)**	Student controls one or more variables. Calculates result independently. Formula hidden after Stage 1. Immediate feedback per submission.	60%
Stage 3	**Challenge (Independent)**	Student given a target outcome and must adjust multiple variables to achieve it. No formula shown. Countdown timer active. Multiple correct combinations possible.	70%

Table 4: Stage Design Summary

Firebase-Driven Difficulty Scaling

Six game parameters are adjusted based on the student's grade level. All parameter values are stored as Firestore documents and loaded into Unity at runtime via the Python FastAPI backend and Flutter, requiring no Unity code changes to support a new grade or topic configuration.

Parameter	Lower Grades (6-7)	Upper Grades (10-11)
Numerical value ranges	Small whole numbers only	Decimal values and larger magnitudes
Number of controlled variables	One variable at a time	Two or more variables simultaneously
Formula visibility in Stage 2	Formula partially visible	Formula completely hidden
Timer duration in Stage 3	Extended (90 seconds)	Compressed (45 seconds)
Hint availability	Hints available throughout	Hints limited to Stage 1 only
Step-by-step explanation depth	Single formula application	Multi-step derivation with unit analysis

Table 5:Difficulty Scaling Parameters by Grade

End-to-End Implementation Flow

The complete data flow from student action to score storage follows six steps. Flutter sends the student's grade and lesson topic to the Python FastAPI backend. Python queries Firestore and returns the matching game configuration JSON. Flutter launches the appropriate Unity scene via flutter_unity_widget and passes the configuration via postMessage(). Unity parses the configuration, initialises the Stage 1 scene with the correct parameter values, and runs physics calculation validation on each student submission. On stage completion, Unity sends a result JSON object to Flutter via sendMessageToFlutter(). Flutter posts the score and measurement data to Firebase Firestore, where Wijerathna's learning performance evaluation module reads it for mastery level calculation.

3.4 Evaluation Strategy

Evaluation is conducted across four dimensions. Retrieval quality evaluation applies Precision@K, Recall@K, and MRR metrics to both BM25 and SBERT on a held-out set of NIE query-document pairs not used in training. Game functional testing verifies measurement accuracy, calculation correctness, and stage unlock validity. Grade-difficulty alignment is assessed by the researcher and supervisor using a rubric against the NIE curriculum for each grade-topic configuration. Where student participants can be recruited for a pilot session, usability is assessed using the System Usability Scale questionnaire. The following benchmarks are intended targets subject to experimental outcomes:

Evaluation Dimension	Planned Metric	Intended Target
BM25 retrieval precision	Precision@5 on held-out query set	Precision@5 >= 0.70
SBERT retrieval improvement over BM25	MRR comparison on indirect query subset	SBERT MRR >= BM25 MRR on paraphrased queries
Grade-filter accuracy	Proportion of results correctly grade-scoped	>= 95% correctly filtered
Game measurement accuracy	Correctness of physics values generated	100% of physics calculations validated correct
Stage unlock validity	Correct threshold enforcement	Zero incorrect stage unlocks in testing
Grade-difficulty alignment	Rubric score by supervisor	>= 80% of configs rated aligned to NIE grade

Table 6:Planned Evaluation Metrics and Intended Benchmarks

3.5 Technology Stack

- Python FastAPI — backend server handling retrieval queries and game config delivery
- rank_bm25 — BM25 Tier 1 keyword retrieval library
- sentence-transformers — SBERT fine-tuning and Tier 2 semantic retrieval
- HuggingFace all-MiniLM-L6-v2 — lightweight base model for SBERT (80MB)
- Google Colab free tier — SBERT fine-tuning training environment
- Firebase Firestore — NIE content storage, game configuration documents, score storage
- Unity 2D — game scene development and physics calculation engine
- flutter_unity_widget — Unity scene embedding within Flutter mobile application

- Flutter (Dart) — mobile application framework and API client

3.6 Ethical Considerations

All training data is sourced from publicly available datasets (SciQ) and official NIE textbooks freely accessible at nie.lk. No private or personally identifiable student data is collected or processed during development. Student participation in pilot usability testing will be voluntary, and no performance data beyond anonymised session records will be retained for research purposes. Unity game scenes will be tested in an isolated development environment before deployment.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Project Structure and Component Scope

The overall PhysiV-Lab project consists of four team-level components: (1) Grade-Adaptive RAG-Based AI Chatbot and Structured Learning Navigation (Sachintha Nirmal S.M.S, IT22234834), (2) Intelligent Keyword-Based Content Retrieval and Curriculum-Aligned 2D Game Generation (this component, Arthanayake C.T.H, IT22251800), (3) Adaptive User Management and Learning Performance Evaluation (Wijerathna C.D.W, IT22123268), and (4) Interactive Physics Experimentation and Learning (Vidurangi H.I.D.G, IT22143990). This proposal describes component 2 only. Within this component, two independent sub-systems are implemented sharing a common FastAPI backend and Firestore database.

4.2 Component Position in Lesson Workflow

Within the overall lesson workflow, this component provides two services at different points. The Content Retrieval sub-system is accessible to the student throughout the lesson as a reference tool when they encounter an unfamiliar term, complementing Sachintha's AI chatbot which handles conversational lesson navigation. The 2D Game sub-system activates as a standalone lesson activity after the student completes Vidurangi's interactive physics experiments, providing a measurement-driven reinforcement experience before Wijerathna's MCQ assessment. Game scores and physics measurement data are forwarded to Wijerathna's learning analytics module for mastery level calculation.

4.3 System Architecture Diagram

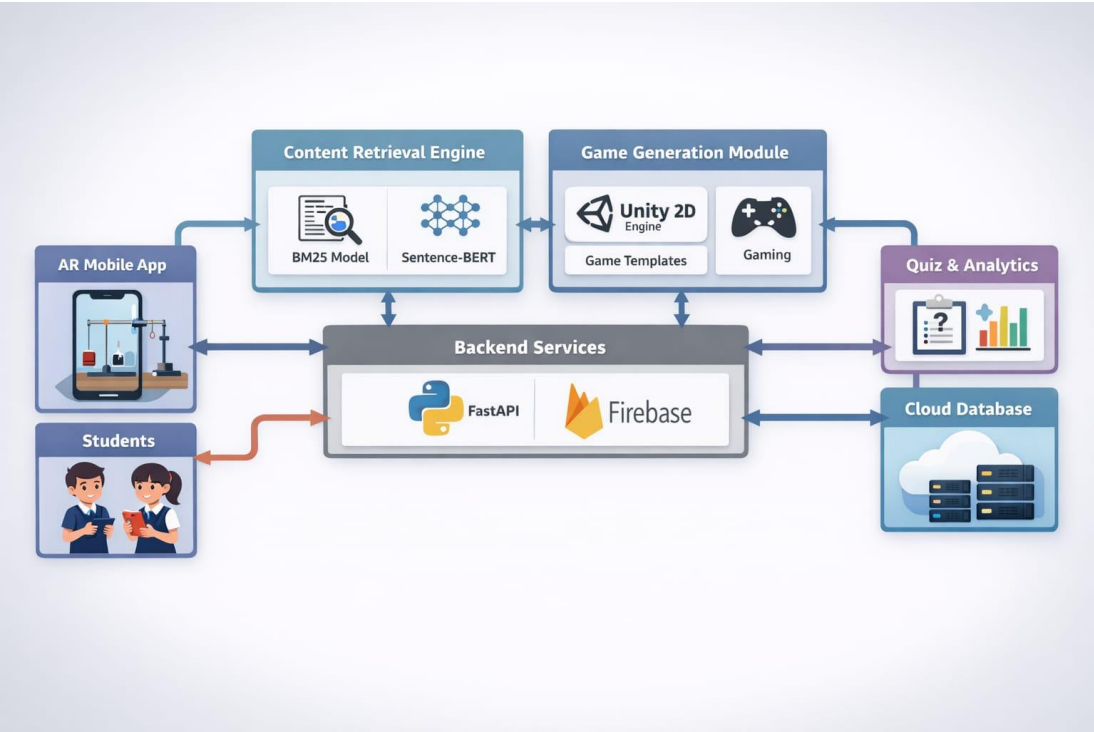


Figure 1:System Architecture Diagram

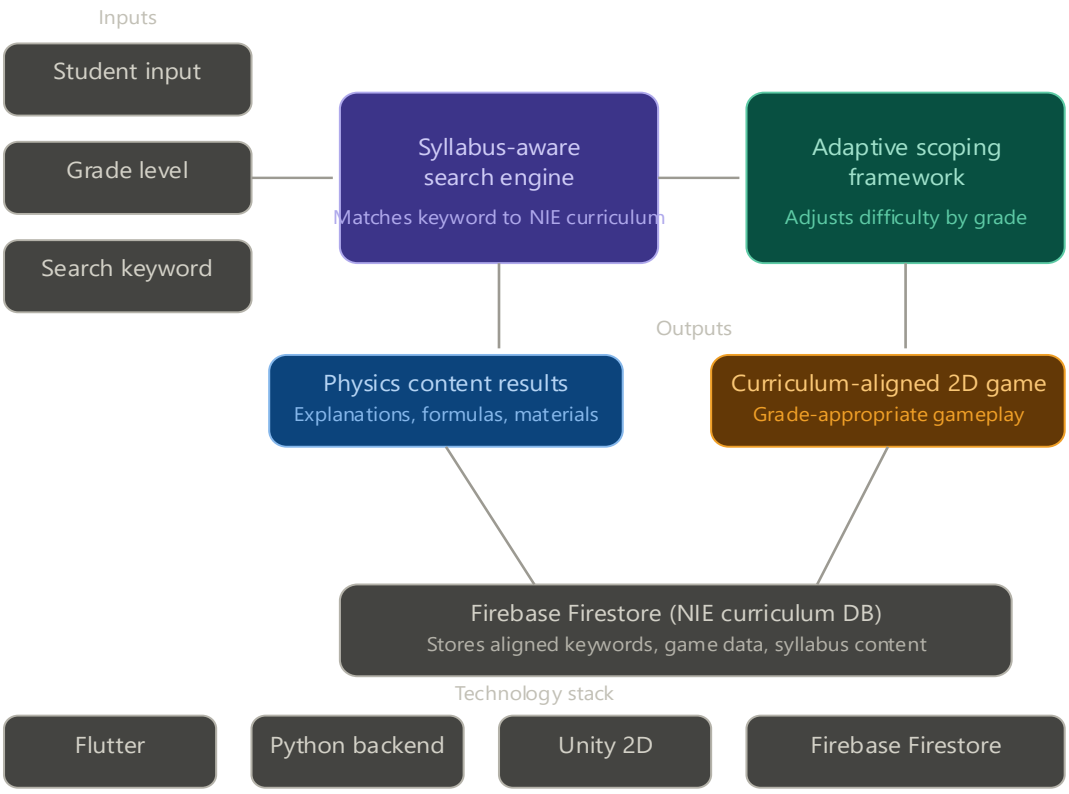


Figure 2: System flow.

4.4 Data Flow

For the retrieval sub-system, the student submits a keyword or natural language query through the Flutter search interface. Flutter sends the query and the student's grade level to the FastAPI POST /retrieve endpoint. The backend applies grade filtering, runs BM25 Tier 1, and invokes SBERT Tier 2 if BM25 confidence is insufficient. The top-ranked result is returned as a JSON object containing the explanation, formula, and example, which Flutter renders on screen.

For the game sub-system, Flutter sends the student's grade and lesson topic to the FastAPI GET /game-config endpoint. The backend queries Firestore and returns the matching game configuration JSON. Flutter launches the Unity scene via flutter_unity_widget, passes the configuration, and receives result JSON from Unity on stage completion. Flutter posts the score and measurement data to Firestore for Wijerathna's module to consume.

4.5 Scalability and Extensibility

The retrieval system is extensible to additional grades, subjects, or curricula by adding new documents to the NIE content Firestore collection. The SBERT model can be retrained on extended datasets without changes to the backend API. The game system is extensible to new topics and grade levels through Firestore configuration documents only, with no Unity code modifications required. The template-based architecture allows new game scenes to be added independently of existing templates.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role	Primary Interest
Grade 6-11 Students (Rural)	Primary	Accessible keyword search for physics concepts and engaging curriculum-aligned game activities
Physics Teachers	Secondary	NIE curriculum alignment of retrieved content and game difficulty, accurate formula representations
System Integration Team	Primary	Correct API data formatting and score forwarding to analytics module
Research Supervisors	Academic	Validated retrieval metrics, measurable game evaluation outcomes, evidence of novelty
Educational Authorities	Secondary	NIE curriculum accuracy and potential adoption for national digital learning initiatives

Table 7:Stakholder identification

5.2 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall accept keyword or natural language queries from students through the Flutter mobile interface.	Must Have
FR-02	The system shall filter retrieval candidates by the student's grade level before ranking.	Must Have
FR-03	The system shall apply BM25 Tier 1 retrieval and return results when confidence exceeds the defined threshold.	Must Have
FR-04	The system shall apply SBERT Tier 2 semantic retrieval when BM25 confidence is below threshold.	Must Have
FR-05	Retrieved results shall include the physics explanation, formula where applicable, and a worked example.	Must Have
FR-06	The system shall load game configuration from Firestore based on the student's grade and lesson topic.	Must Have
FR-07	The system shall launch the appropriate Unity game template and pass configuration via postMessage().	Must Have
FR-08	Unity game scenes shall implement three progressive stages with mastery-based unlock thresholds.	Must Have
FR-09	Each student game interaction shall produce a real-time physics measurement validated against the correct formula.	Must Have
FR-10	The system shall display step-by-step formula explanations derived from measured values.	Must Have
FR-11	Stage difficulty parameters shall be driven by Firestore configuration without Unity code modification.	Must Have
FR-12	Game scores and measurement data shall be forwarded to Firebase for the analytics module to consume.	Must Have

FR-13	The system shall prevent stage advancement when the student has not met the pass threshold.	Must Have
FR-14	The retrieval system shall return a result within an acceptable response time.	Should Have

Table 8:Functional Requirements

5.3 Non-Functional Requirements

ID	Requirement	Intended Target
NFR-01	The retrieval system shall return a ranked result within an acceptable response time on the backend.	Response within 5 seconds per query
NFR-02	The SBERT model shall run inference on CPU without GPU dependency.	CPU-only inference on FastAPI backend
NFR-03	The game system shall load configuration and launch the Unity scene within an acceptable time.	Scene loaded within 6 seconds on Android
NFR-04	Physics calculation validation in Unity shall produce zero incorrect results.	100% calculation accuracy in functional testing
NFR-05	New game topics and grades shall be addable through Firestore configuration only.	Configuration-only addition verified by test
NFR-06	The system shall handle missing or invalid Firestore game configurations gracefully.	Zero unhandled exceptions in testing
NFR-07	All NIE content retrieved shall be scoped to the student's grade level.	>= 95% grade-filter accuracy

Table 9: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market is Grade 6-11 students in rural and semi-urban government schools in Sri Lanka who own Android smartphones but lack access to curriculum-aligned digital reference materials and physics laboratory facilities. Secondary markets include private tutoring centers seeking adaptive physics practice tools, educational technology publishers requiring curriculum-aligned game content for the Sri Lankan market, and government education authorities interested in scalable NIE-aligned digital learning solutions for national examination preparation support.

6.2 Value Proposition

The proposed component provides two distinct value propositions unavailable in any existing platform. For students, it is the only mobile physics tool that provides grade-filtered semantic keyword search aligned to the NIE syllabus, ensuring that retrieved explanations and formulas match the exact curriculum level of the student. For physics learning through games, it is the only platform that converts gameplay actions into real measured physics values, enforces mastery-based stage progression, and scales difficulty automatically to the NIE grade level without requiring curriculum experts to modify game code.

6.3 Revenue Model

Should the PhysiV-Lab platform be commercialized, this component supports a freemium model where core retrieval and basic game templates are available at no cost. Premium features including expanded game template libraries, additional NIE topic configurations, teacher-facing analytics on student game performance, and advanced SBERT retrieval could be offered through institutional subscriptions or government licensing arrangements.

6.4 Competitive Advantage

No existing mobile physics platform for Sri Lankan school students provides NIE-curriculum-aligned semantic search with grade filtering. The template-based Firebase-driven game architecture means new curriculum topics can be added without software development, significantly reducing the marginal cost of content expansion compared to custom-built educational games. The measurement-driven game design produces richer learning analytics data than score-only game platforms, providing a distinct advantage for integration with learning performance evaluation systems.

6.5 Intellectual Property Considerations

The NIE physics content dataset and curriculum-aligned query-document pairs constructed for this research represent original intellectual contributions. The two-tier BM25 plus NIE-fine-tuned SBERT retrieval architecture applied to school-level physics curriculum search may represent a novel contribution if validated. The Firebase-driven grade-scoping game architecture may be protectable as a software design pattern. Appropriate copyright and licensing will be established in consultation with the university's commercialization guidelines following academic publication.

7. BUDGET AND JUSTIFICATION

Category	Item		Qty	Unit (LKR)	Total (LKR)
Software	All planned software (Python, rank_bm25, sentence-transformers, HuggingFace, Unity Personal, Flutter, Firebase SDK) — all open-source or free tier		—	0	0
Cloud Services	Firebase Firestore free tier — sufficient for NIE content dataset and game configs		—	0	0
Compute	SBERT fine-tuning on Google Colab free tier — no GPU purchase needed		—	0	0
Hardware	Development and inference on researcher's personal laptop and Android test device		—	0	0
Contingency	Approximately 10 percent buffer		—	—	0
TOTAL					**LKR 0**

Table 10:Budget summery

Justification

- All software including Python, rank_bm25, sentence-transformers, HuggingFace model weights, Unity Personal edition, Flutter, and Firebase SDK are available under open-source or free-use licenses at zero cost.
- SBERT model fine-tuning is conducted on Google Colab free tier, which provides sufficient GPU compute for a dataset of 200 to 400 training pairs without purchase.
- Firebase Firestore free tier provides sufficient read and write capacity for the NIE content dataset and game configuration documents at the prototype scale.
- The NIE physics content dataset is constructed from publicly available NIE textbooks at nie.lk, incurring no data acquisition cost.
- Unity Personal edition is available at no cost for non-commercial research projects meeting the revenue threshold, which this academic prototype satisfies.

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Planned Activity	Planned Deliverable	Est. Duration
1.0	**Requirements and Design**		
1.1	Define NIE content schema and target concept list for all six grades	Content schema specification	3 days
1.2	Design retrieval API interface and game config schema	API and schema documents	3 days
1.3	Define game template designs and Firebase config structure	Game design document	4 days
2.0	**Dataset and Artefact Construction**		
2.1	Extract and structure NIE physics content from textbooks (Grades 6-11)	200-400 entry NIE dataset in Firestore	2 weeks
2.2	Author query strings for each NIE dataset document	Complete query-document pairs	1 week
3.0	**Sub-System 1: Retrieval Development**		
3.1	Implement BM25 retrieval with grade filtering and evaluate as baseline	BM25 baseline with evaluation results	1 week
3.2	Fine-tune SBERT on SciQ for domain adaptation (Google Colab)	Domain-adapted SBERT model	1 week
3.3	Fine-tune SBERT on NIE query-document pairs	NIE-specialised SBERT model	1 week
3.4	Implement cascaded Tier 1 / Tier 2 retrieval logic in FastAPI	Working two-tier retrieval API	4 days
3.5	Evaluate retrieval system on held-out NIE query set	Retrieval evaluation report	4 days
4.0	**Sub-System 2: Game Development**		
4.1	Build Unity Linear Motion Lab template with stage logic and calculation validation	Working game template 1	1.5 weeks
4.2	Build Unity Force and Mass Arena template	Working game template 2	1 week
4.3	Build Unity Pressure Chamber template	Working game template 3	1 week

4.4	Implement Firebase config loading and grade-parameter injection in Unity	Firebase-driven config system	4 days
4.5	Implement flutter_unity_widget integration and result forwarding to Firestore	Flutter-Unity integration	4 days
5.0	**Integration and Testing**		
5.1	Integrate both sub-systems with FastAPI backend and Flutter app	Integrated prototype	1 week
5.2	Functional testing of retrieval pipeline and game measurement accuracy	Functional test report	1 week
5.3	Usability pilot with student participants where feasible	Usability evaluation report	1 week
6.0	**Documentation and Reporting**		
6.1	Write research report chapter for this component	Draft report chapter	2 weeks
6.2	Prepare presentation materials	Presentation slides	3 days

Table 11: Work Breakdown Structure

9. GANTT CHART

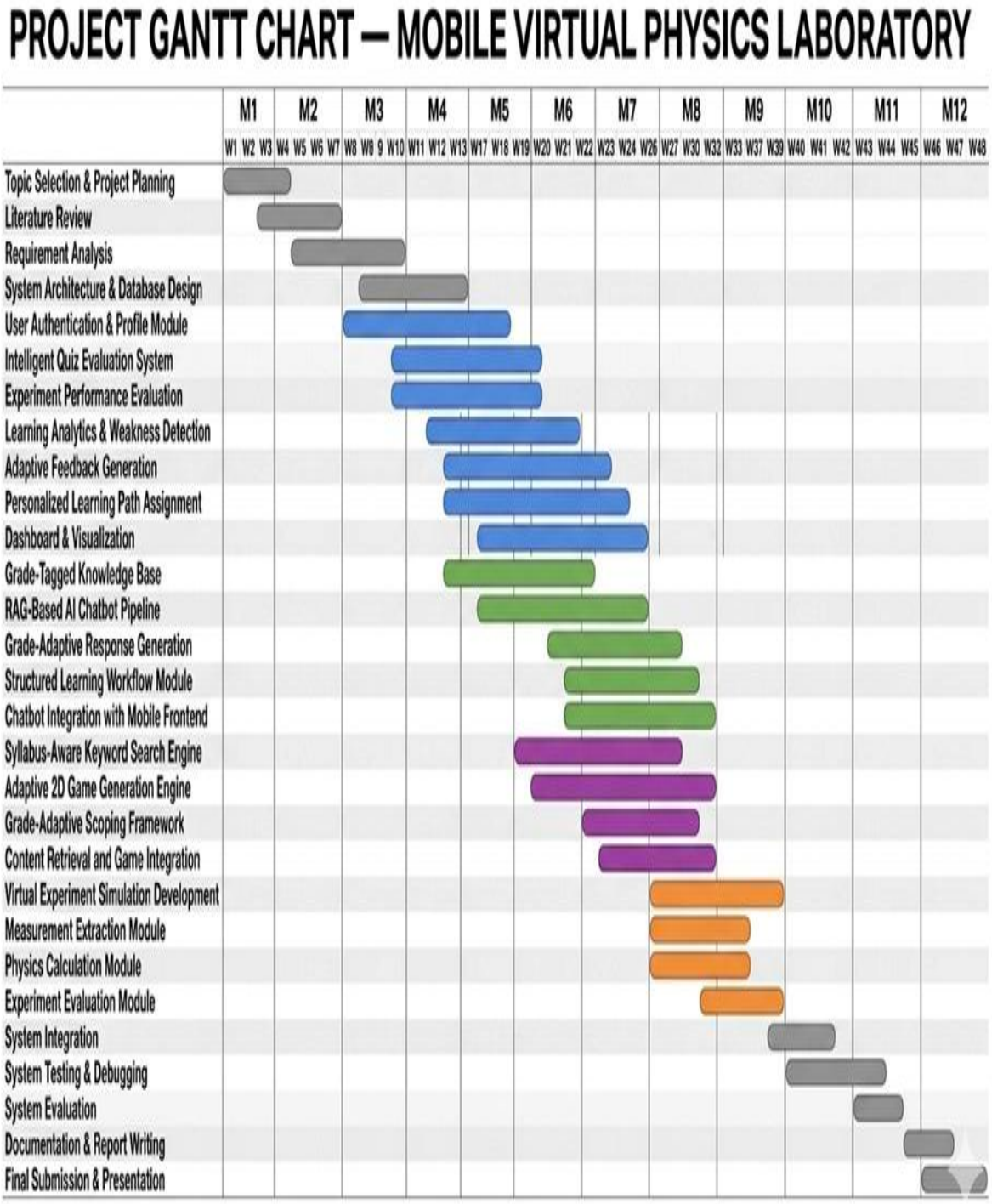


Figure 3: Grant chart

10. REFERENCES LIST

[1] M. Kalogiannakis, S. Papadakis, and A. I. Zourmpakis, "Gamification in Science Education: A Systematic Review of the Literature," Education Sciences, vol. 11, no. 1, pp. 1-36, 2021. doi: 10.3390/educsci11010022

[2] A. B. Gil, A. de la Prieta, S. Rodriguez, and J. M. Corchado, "Smart System for the Retrieval of Digital Educational Content Based on Machine Learning," Applied Sciences, vol. 9, no. 20, 2019. doi: 10.3390/app9204400

[3] R. Vijayatheepan, "Virtual Labs in Science Teaching: Teachers' Perspectives and Challenges in Sri Lankan Schools," Journal of Innovation in Physics Education, Open University of Sri Lanka Repository, 2025.

[4] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in Proc. 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP), Hong Kong, 2019, pp. 3982-3992. doi: 10.18653/v1/D19-1410

[5] R. Elmoazen, M. H. Al-Mamun, and A. Al-Hassan, "Learning Analytics in Virtual Laboratories: A Systematic Literature Review of Empirical Research," Smart Learning Environments, vol. 10, no. 18, 2023. doi: 10.1186/s40561-023-00244-y

[6] M. A. Rahman, M. A. Hossain, and M. S. Islam, "Virtual Laboratories in STEM Education," Journal of Advanced Learning and Teaching, vol. 4, no. 2, pp. 1-15, 2021.

[7] Y. Alotaibi and M. Alotaibi, "A Framework for Integrating Game-Based Learning within Mobile Education Applications," International Journal of Advanced Computer Science and Applications, vol. 12, no. 3, pp. 450-458, 2021. doi: 10.14569/IJACSA.2021.0120352

[8] J. Welbl, N. F. Liu, and M. Gardner, "Crowdsourcing Multiple Choice Science Questions," in Proc. Workshop on Noisy User-generated Text (W-NUT), EMNLP 2017, Copenhagen, Denmark, 2017, pp. 94-106. doi: 10.18653/v1/W17-4413

[9] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design Science in Information Systems Research," MIS Quarterly, vol. 28, no. 1, pp. 75-105, Mar. 2004. doi: 10.2307/25148625

[10] S. E. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," Foundations and Trends in Information Retrieval, vol. 3, no. 4, pp. 333-389, 2009. doi: 10.1561/1500000019

[11] W. Huang, K. F. Hew, and B. Lo, "Investigating the Effects of AI Chatbots on Students' Learning Outcomes and Motivation," Educational Technology & Society, vol. 22, no. 2, pp. 1-13, 2019.

[12] O. Zawacki-Richter, V. I. Marin, M. Bond, and F. Gouverneur, "Systematic Review of Research on Artificial Intelligence Applications in Higher Education," International Journal of Educational Technology in Higher Education, 2019. doi: 10.1186/s41239-019-0171-0

11. APPENDICES

Appendix A: Preliminary Feasibility Study

To assess technical viability, a preliminary investigation was conducted into the availability of NIE physics content and the behaviour of BM25 retrieval on a small manually authored sample. A set of 30 query-document pairs was constructed from Grade 8 NIE physics topics covering speed, force, and pressure. BM25 was applied to these 30 documents using rank_bm25, and retrieval was tested with 15 direct and 10 indirect query strings. Preliminary observations indicate that BM25 achieves strong precision for direct keyword queries (exact concept terms) but misranks several indirect paraphrased queries, motivating the addition of the SBERT Tier 2 semantic fallback. These are preliminary observations based on a small manually curated sample and should not be interpreted as validated final retrieval metrics.

A preliminary Unity prototype was also constructed for the Linear Motion Lab template to verify that the flutter_unity_widget integration is functional on an Android test device and that the postMessage and sendMessageToFlutter communication channels operate correctly. The prototype confirms that game configuration data can be injected from Flutter into Unity at runtime and that physics calculation results can be forwarded back to Flutter for Firebase storage. This supports the technical feasibility of the template-based Firebase-driven game architecture.

Appendix B: NIE Physics Content Dataset Sample Entries

The following illustrates the planned structure and content for two sample entries from the NIE physics content dataset:

Field	Grade	Content
concept_id	8	speed_grade8
topic	8	Speed, Distance and Time
explanation	8	Speed is how fast an object moves. It is calculated by dividing the distance travelled by the time taken. If a car travels 120 km in 2 hours, its speed is 60 km/h.
formula	8	speed = distance / time (v = s / t) where v is speed in m/s, s is distance in metres, t is time in seconds
example	8	A cyclist covers 500 m in 25 seconds. Speed = 500 / 25 = 20 m/s.
keywords	8	speed, distance, time, velocity, fast, slow, calculation, formula

query_strings	8	['how do you calculate speed', 'what is the formula for speed', 'speed distance time', 'how fast is something moving']

Equation 1:NIE physics content dataset

Appendix C: Firebase Game Configuration Schema Example

The following shows the Grade 8 Speed configuration document stored in Firestore for the Linear Motion Lab game template:

Field	Value (Grade 8 Speed)
template_id	linear_motion_lab
grade	8
topic	Speed
formula	v = s / t
stage1_distance_range	[100, 500] metres (whole numbers)
stage1_time_range	[10, 50] seconds (whole numbers)
stage2_controlled_var	distance
stage2_formula_visible	false
stage2_pass_threshold	0.60
stage3_target_speed	randomly selected from [15, 20, 25] m/s
stage3_timer_seconds	60
stage3_pass_threshold	0.70
hint_available_stages	[1, 2]

Equation 2:Linear Motion Lab game template

Appendix D: Risk Analysis

Risk	Likelihood	Impact	Mitigation
NIE textbook content insufficient for SBERT fine-tuning (< 200 pairs)	Low	High	*Supplement with SciQ and augment NIE pairs through paraphrase generation*

SBERT fine-tuned model shows no improvement over BM25 baseline	Medium	Medium	Report BM25 as primary system; present SBERT as exploratory contribution with analysis of failure cases
flutter_unity_widget incompatibility on target Android version	Low	High	Test on minimum API 21 device early; maintain WebView-based fallback game if Unity integration fails
Google Colab session timeout during SBERT training	Medium	Low	Save model checkpoints after each epoch; use Colab Pro free trial if needed
Firebase Firestore free tier quota exceeded during testing	Low	Low	Cache Firestore reads locally in Flutter; batch writes to reduce quota usage
Insufficient student participants for usability pilot	Medium	Medium	Conduct evaluation with researcher, supervisor, and team members as proxy users; supplement with rubric assessment

Table 12:Risk Analysis

Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students

R26-IT-150

Project Proposal Report

Sachintha Nirmal S M – IT22234834

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

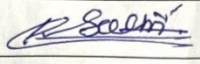
Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Sachintha Nirmal S M	IT22234834	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Indeewari Dahanayake	18/03/2026	

Scanned with CamScanner

ABSTRACT (Maximum 300 words)

Rural school students in Sri Lanka face significant barriers to quality physics education, particularly the absence of laboratory facilities and limited access to interactive learning support. Although digital platforms have emerged to address access to experimentation, few systems integrate real-world contextual learning with intelligent student support at the lesson level. Existing virtual physics platforms provide either simulation activities or static quiz assessments independently, without connecting real-world events to curriculum topics or offering an intelligent mechanism for students to resolve conceptual confusion after assessment.

This research proposes a two-part intelligent learning support component within the PhysiV-Lab mobile virtual physics laboratory system, targeting Grade 6-11 students in rural Sri Lanka. The first part is a Real-World News-Based Scenario Question Generator, which automatically collects news articles from curated international sources via RSS, classifies physics-relevant content using zero-shot NLP, maps events to grade-appropriate curriculum topics, and generates contextual scenario-based questions using a locally deployed large language model. The second part is an AI-Powered Post-Lesson Q&A Agent, which activates after lesson MCQ completion and accepts free-text student questions in English or Sinhala, automatically extracts physics keywords using NLP, retrieves grade-filtered explanations via BM25 ranking over a structured physics knowledge base, and falls back to LLM-generated responses when the knowledge base does not contain a match.

Technologies used include Python FastAPI for the backend, feedparser for RSS collection, facebook/bart-large-mnli for zero-shot classification, Ollama with Llama 3.1 for local LLM inference, spaCy and rank_bm25 for keyword extraction and retrieval, Helsinki-NLP models for Sinhala-English translation, and Flutter for mobile interface development. All processing uses open-source tools at zero cost, with the knowledge base and curriculum mapping constructed as original research artefacts from the Sri Lankan physics curriculum.

Expected outcomes include a deployed end-of-lesson scenario question pipeline grounded in current global events, and a responsive multilingual Q&A agent that reduces student conceptual confusion before the scenario challenge. Together these components form a closed real-world contextual learning loop that advances physics education accessibility for rural learners. The system supports SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities).

Keywords: Real-World Contextual Learning, Scenario-Based Assessment, Retrieval-Augmented Generation, BM25 Information Retrieval, Multilingual AI Agent, Mobile Physics Education

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor, Mr. Indunil Daluwatte, for his continuous guidance, encouragement, and invaluable feedback throughout the development of this research proposal. His expertise and dedication have greatly shaped the direction and quality of this work.

I extend my appreciation to the academic staff of the Department of Information Technology at the Sri Lanka Institute of Information Technology for providing the knowledge, resources, and academic environment necessary to carry out this research. I am also grateful to my fellow project team members for their collaboration and constructive discussions during the design of the overall system architecture.

Finally, I wish to express my deepest gratitude to my family and friends for their unwavering support, patience, and encouragement throughout my academic journey.

TABLE OF CONTENT (Auto Generated)

Contents

DECLARATION 2
ABSTRACT (Maximum 300 words) 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT (Auto Generated) 5
LIST OF TABLES 7
LIST OF FIGURES 7
LIST OF ABBREVIATIONS 7
LIST OF APPENDICES 7
1. INTRODUCTION 8
1.1 Background and Context 8
1.2 Literature Review 9
1.3 Research Gap 10
2. OBJECTIVE 12
2.1 Main Objective 12
2.2 Specific Objectives 12
3. METHODOLOGY 13
3.1 Research Approach 13
3.2 Part One: News-Based Scenario Question Pipeline 13
3.2.1 News Collection 13
3.2.2 Physics Relevance Classification 13
3.2.3 Curriculum Mapping 13
3.2.4 Scenario Question Generation 13
3.2.5 Question Delivery 14
3.3 Part Two: Post-Lesson AI Q&A Agent 14
3.3.1 Lesson Trigger and Q&A Session 14
3.3.2 Language Detection and Translation 14
3.3.3 Keyword Extraction 14
3.3.4 BM25 Retrieval with Grade Filtering 14
3.3.5 LLM Fallback 15
3.4 Knowledge Base Construction 15
3.5 Evaluation Strategy 15
3.6 Ethical Considerations 16
4. HIGH-LEVEL SYSTEM ARCHITECTURE 17
4.1 Project Structure and Component Scope 17

4.2 Component Position in Lesson Workflow 17
4.3 Data Flow 18
4.4 Scalability and Extensibility 18
5. USER REQUIREMENTS 19
5.1 Stakeholder Identification 19
5.2 Functional Requirements 19
5.3 Non-Functional Requirements 20
6. COMMERCIALIZATION PLAN 21
6.1 Target Market and Customer Persona 21
6.2 Value Proposition 21
6.3 Revenue Model 21
6.4 Competitive Advantage 21
6.5 Intellectual Property Considerations 21
7. BUDGET AND JUSTIFICATION 22
Justification 22
8. WORK BREAKDOWN STRUCTURE (WBS) 23
9. GANTT CHART 25
10. REFERENCES LIST 26
11. APPENDICES 27
Appendix A: Preliminary Feasibility Study 27
Appendix B: Physics Knowledge Base Sample Entries 27
Appendix C: Risk Analysis 27
Appendix D: Curriculum Mapping Table Sample 28

LIST OF TABLES

Table 1:List_Of_Abbreviations 7
Table 2: Research_Gap 11
Table 3: Planned Evaluation Metrics and Intended Benchmarks 16
Table 4:Stakeholder_Indentification 19
Table 5:Functional_Requirements 20
Table 6:Non_Functional_Requirments 20
Table 7:Budget_Summary 22
Table 8:Work_Breakdown_Structure 24
Table 9:Explanation_01 27
Table 10:Explanation_02 28
Table 11: Explanation_03 28

LIST OF FIGURES

Figure 1:System_Architecture 17
Figure 2:Gantt_chart 25

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BM25	Best Match 25 (Information Retrieval Algorithm)
LLM	Large Language Model
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
RSS	Really Simple Syndication
SDG	Sustainable Development Goal
Q&A	Question and Answer
MCQ	Multiple Choice Question

Table 1:List_Of_Abbreviations

LIST OF APPENDICES

Appendix A: Preliminary Feasibility Study (Page 27)

Appendix B: Physics Knowledge Base Sample Entries (Page 27)

Appendix C: Risk Analysis (Page 28)

Appendix D: Curriculum Mapping Table Sample (Page 28)

1. INTRODUCTION

1.1 Background and Context

Real-World Problem

Physics education in rural Sri Lanka is severely constrained by the absence of laboratory infrastructure, limited instructional support, and a persistent gap between theoretical content and practical application [1], [2]. While several virtual laboratory and mobile learning platforms have been proposed to address access to experimentation, most of these systems confine students to static simulations and predefined quiz assessments that are disconnected from the wider world students inhabit. This creates a conceptual wall between physics as a curriculum subject and physics as the explanation for observable global events.

A particularly underexplored limitation in existing educational technology is the absence of mechanisms that link real-world current events to physics curriculum content in a grade-adapted, automated, and contextually meaningful way. When major events occur globally, such as missile launches, space missions, earthquake disasters, or engineering accidents, these events contain rich physics content that could be leveraged to situate classroom knowledge in lived reality. No existing mobile physics learning platform for school-level students in developing contexts has operationalized this connection.

Simultaneously, students completing lesson assessments frequently encounter conceptual confusion that they cannot resolve independently. Existing systems either provide no post-assessment support, or offer generic AI chatbot interaction that is not grounded in the specific curriculum topic just studied. A need therefore exists for an intelligent, responsive, and multilingual post-lesson support mechanism that operates within the lesson workflow rather than as a separate application.

Who is Affected and Why It Matters

The population most significantly affected includes approximately 1.2 million students enrolled in Grades 6 through 11 in government schools across Sri Lanka, a significant proportion of whom are in rural or semi-urban areas with limited laboratory access [2]. Physics consistently registers among the weakest performance areas in national examinations, with rural students disproportionately affected. The inability to connect theoretical concepts to real-world phenomena is cited by educators as a key contributor to low engagement and retention in physics [1], [3]

Statistical Evidence

Research by Vijayatheepan [2] documents that virtual laboratory access is unavailable in the majority of rural Sri Lankan schools, while Rahman et al. [1] demonstrate that virtual experiments can improve conceptual understanding by up to 34 percent compared to lecture-only instruction. Studies in contextual learning consistently show that grounding abstract scientific content in real-world events improves both motivation and retention, particularly for secondary school learners [4], [6]. Despite this evidence, no mobile platform for Sri Lankan school physics currently implements automated real-world contextual question generation.

Analysis of Existing Solutions and Their Limitations

Existing virtual physics platforms such as PhET Interactive Simulations and Labster provide interactive experiment environments but do not include contextual question generation from live news or post-lesson AI-assisted Q&A [3]. Gamified learning platforms such as Kahoot and Quizziz offer engaging quiz formats but are not curriculum-aligned, do not adapt to grade level, and do not incorporate real-world scenario content. Intelligent tutoring systems and AI chatbots such as Khanmigo provide general subject assistance but are not integrated within a structured lesson workflow, do not support Sinhala, and are not designed for low-resource mobile environments [7], [8].

Most critically, no existing system combines automated news-driven contextual question generation with a post-lesson multilingual retrieval-augmented Q&A agent operating within a single mobile lesson workflow. This dual gap is the focus of the proposed research component

SDG Alignment

This component supports SDG 4 (Quality Education) by improving access to contextually rich and intelligently supported physics learning for underserved rural students. It supports SDG 10 (Reduced Inequalities) by delivering equal access to intelligent educational support regardless of geographic location. The research is aligned with the Technology Integration and Management (TIM) research cluster and the Information Technology (IT) specialization.

1.2 Literature Review

A review of ten recent scholarly sources was conducted to examine the current state of research in virtual laboratory education, contextual and scenario-based learning, intelligent content retrieval, and multilingual AI support in educational systems.

Rahman et al. [1] and Vijayatheepan [2] establish the foundational evidence for virtual laboratory effectiveness in developing-country school contexts, demonstrating measurable improvements in conceptual understanding while identifying persistent infrastructure limitations in rural Sri Lanka. Elmoazen et al. [3] provide a systematic review of learning analytics in virtual laboratory environments and conclude that most existing platforms focus on activity logs and experiment interaction without integrating contextual or scenario-based assessment at the lesson level.

In the domain of contextual and scenario-based learning, Kalogiannakis et al. [4] demonstrate through systematic review that gamification and contextual engagement techniques significantly improve science learning motivation and retention in secondary education. Their analysis shows that connecting abstract physics concepts to observable or current-events-based scenarios produces stronger conceptual transfer than isolated assessment. However, the reviewed systems do not implement automated mechanisms for generating such contexts from live data sources.

For information retrieval in educational systems, Gil et al. [5] propose a smart retrieval system for digital educational content using machine learning and demonstrate that keyword-based retrieval with ranking algorithms can effectively match student queries to relevant educational materials. Their architecture, however, is designed as a standalone system rather than an integrated module within a lesson workflow, and does not support grade-filtered retrieval or multilingual input.

In the area of AI-powered educational support, Huang et al. [6] investigate the effects of AI chatbots on student learning outcomes and motivation, finding significant improvements in student engagement when AI assistance is available during or immediately after learning activities. Zawacki-Richter et al. [7] further confirm through systematic review that AI applications in education are most effective when tightly integrated with structured learning workflows rather than deployed as separate tools. Neither study examines retrieval-augmented generation applied to post-lesson physics Q&A or multilingual support for South Asian school learners.

Lewis et al. [9] introduced the concept of retrieval-augmented generation (RAG), demonstrating that combining a retrieval system with a generative language model substantially improves factual accuracy and relevance compared to generation alone. This approach is directly applicable to the proposed Q&A agent, where BM25 retrieval over a curated knowledge base provides grounded answers before LLM generation is used as a fallback. Robertson and Zaragoza [10] formally established the BM25 ranking algorithm as the standard in probabilistic information retrieval, providing the theoretical basis for the keyword-ranked search mechanism in the proposed system.

Existing commercial and open-source educational products further illustrate the gap. PhET simulations [3] and Labster provide excellent experiment environments but offer no contextual question generation or post-lesson Q&A. Khanmigo and similar AI tutors provide conversational physics support but operate independently of lesson workflows, do not support Sinhala, and require persistent internet connectivity inappropriate for rural deployments [7], [8]. RSS-based news aggregation is well established in information systems research as a reliable, low-cost mechanism for structured content collection [11], while zero-shot text classification using transformer models has been validated as an effective approach for topic filtering without training data [12]

1.3 Research Gap

Despite advances in virtual laboratory education, contextual learning research, and AI-powered educational tools, a clear gap exists in the integration of these capabilities within a single mobile physics lesson workflow for rural developing-country learners.

Most existing physics education platforms do not incorporate real-world current events as sources of assessment content. Where scenario-based questions exist in educational research, they are authored manually and remain static, failing to reflect the currency of global events relevant to physics concepts. No system reviewed automatically extracts physics-relevant content from live news sources, maps it to grade-level curriculum topics, and generates adapted scenario questions through a local LLM pipeline.

Post-assessment support in existing systems is either absent or provided through generic AI chatbots that are not grounded in curriculum content, do not operate within the lesson workflow, and do not support the native languages of rural Sri Lankan learners. The combination of automatic keyword extraction, grade-filtered BM25 retrieval over a structured physics knowledge base, and multilingual support with LLM fallback has not been proposed or evaluated in a mobile physics education context.

Dimension	Gap in Existing Literature	This Study Aims To Investigate
Scenario question source	Questions are static or manually authored; no system generates from live news	Whether automated news-driven scenario question generation is feasible and curriculum-aligned
Grade adaptation	Contextual questions are not adapted to grade level or curriculum mapping	Whether grade-filtered news mapping and LLM prompting produce appropriately levelled scenario questions
Post-lesson support	Post-assessment AI support is separate from lesson workflow and not curriculum-specific	Whether an in-workflow AI Q&A agent improves conceptual clarity before the scenario challenge
Multilingual support	Existing AI tools do not support Sinhala input in school physics contexts	Whether automatic language detection and translation enables effective Q&A for Sinhala-speaking learners
Retrieval grounding	AI chatbots generate answers without curriculum-grounded retrieval	Whether BM25 retrieval over a structured knowledge base improves answer relevance over LLM-only generation
Knowledge base artefact	No structured grade-segmented physics knowledge base exists for Sri Lankan secondary curriculum	Whether a manually curated grade-segmented knowledge base can serve as an effective retrieval corpus

Table 2: Research_Gap

2. **OBJECTIVE**

2.1 Main Objective

To design, implement, and evaluate an intelligent real-world contextual learning support component that automatically generates grade-adaptive physics scenario questions from curated news sources and provides a multilingual AI-powered post-lesson Q&A agent with curriculum-grounded retrieval, within a mobile virtual physics laboratory application for Grade 6-11 students in rural Sri Lanka

2.2 Specific Objectives

1. To design and implement an automated RSS-based news collection pipeline that periodically ingests articles from a defined set of curated international news sources and stores structured article content for downstream processing.

2. To design and evaluate a physics relevance classification mechanism using zero-shot NLP that filters collected news articles and identifies those containing content mappable to Sri Lankan Grade 6-11 physics curriculum topics.

3. To design and implement a curriculum mapping artefact and grade-filtering mechanism that assigns physics-relevant news articles to appropriate grade bands and lesson topics within the national physics syllabus.

4. To implement and evaluate a local LLM-based scenario question generation pipeline that produces grade-adapted, curriculum-aligned multiple choice questions from physics-relevant news content, with structured JSON output stored for delivery.

5. To design and implement an AI-powered post-lesson Q&A agent that accepts free-text student questions in English or Sinhala, extracts physics keywords automatically, retrieves grade-filtered explanations from a structured knowledge base using BM25 ranking, and generates fallback responses using a local LLM when retrieval does not produce a confident match.

6. To construct and evaluate a structured grade-segmented physics knowledge base from Sri Lankan secondary curriculum materials as the primary retrieval corpus for the Q&A agent.

3. METHODOLOGY

3.1 Research Approach

This study adopts a Design Science Research methodology, which is well established in Information Systems research for designing, implementing, and evaluating software artefacts that address identified practical problems [13]. The study will proceed through four phases: problem identification and requirements analysis, system design and artefact construction, prototype implementation, and experimental evaluation. Evaluation will be conducted through functional testing, pipeline output quality assessment, and usability evaluation with student participants where feasible.

3.2 Part One: News-Based Scenario Question Pipeline

3.2.1 News Collection

News articles will be collected quarterly from a defined set of up to fifteen international news sources including BBC, CNN, Al Jazeera, and Reuters, using their publicly available RSS feeds. The feedparser Python library will be used to parse feed content, and article text will be extracted and stored in Cloud Firestore with metadata including source, publication date, and headline. No paid API access is required. The quarterly collection will be triggered manually during the research phase, with a Python script executing the full pipeline on the researcher's workstation.

3.2.2 Physics Relevance Classification

Collected articles will be classified for physics relevance using zero-shot text classification with the facebook/bart-large-mnli transformer model from HuggingFace. A set of candidate physics concept labels aligned to the Sri Lankan curriculum will be defined, including motion and forces, energy and waves, electricity, pressure and fluids, light and optics, heat and temperature, and not related to physics. Articles receiving a top prediction of not related to physics above a confidence threshold of 0.6 will be discarded. Passing articles will be tagged with their predicted physics concept category for downstream mapping.

3.2.3 Curriculum Mapping

A curriculum mapping artefact will be manually constructed as a structured JSON document that maps each physics concept category to grade bands (Grades 6-7, 8-9, and 10-11) and specific lesson topics within the Sri Lankan national physics syllabus. This artefact constitutes an original research contribution. Articles tagged by the relevance classifier will be matched against this mapping to determine their eligible grade bands and associated lesson topics. Articles with no plausible curriculum mapping will be excluded from question generation.

3.2.4 Scenario Question Generation

For each article passing the curriculum mapping stage, a locally deployed Llama 3.1 8B model via Ollama will generate a grade-adapted multiple choice scenario question. A structured prompt will provide the article context, the target physics concept, the specific lesson topic,

and the target grade band, instructing the model to produce a single four-option MCQ with one correct answer and an explanation, formatted as JSON. Questions will be validated for structural completeness before storage in the Firestore scenario_questions collection, tagged with grade range, lesson topic, news source, article date, and physics concept.

3.2.5 Question Delivery

At the end of the post-lesson Q&A session, the Flutter application queries Firestore for a scenario question matching the student's grade and the lesson topic just completed. If multiple matching questions exist, one is selected. If no question exists for a topic-grade combination, a pre-authored fallback scenario question is served. The student receives one scenario question per lesson as the final challenge before lesson completion is recorded.

3.3 Part Two: Post-Lesson AI Q&A Agent

3.3.1 Lesson Trigger and Q&A Session

The Q&A agent activates only after the student has completed all MCQ questions in Wijerathna's assessment module. A free-text input form is then presented with the prompt Do you have any questions about this lesson? The student may submit as many questions as desired. The session is mandatory before the scenario question is unlocked, but the student may proceed by tapping the I am Ready button at any point after at least one question has been asked and answered, or by choosing to proceed directly if they have no questions.

3.3.2 Language Detection and Translation

On submission of each student question, the langdetect Python library will detect whether the input is in Sinhala or English. If Sinhala is detected, the Helsinki-NLP/opus-mt-si-en transformer model will translate the question to English before processing. After the answer is generated, it will be translated back to Sinhala using the Helsinki-NLP/opus-mt-en-si model before display. This ensures full bidirectional Sinhala support using locally deployed models with no API cost or cloud dependency.

3.3.3 Keyword Extraction

The translated or original English question will be processed by spaCy using the en_core_web_sm model to extract candidate physics keyword phrases through noun phrase and named entity detection. Extracted candidates will then be passed to the Ollama LLM with a focused prompt requesting identification of physics-specific keywords from the candidate list, returning a JSON array. This two-stage approach reduces unnecessary LLM calls for straightforward questions while improving extraction accuracy for complex phrasing.

3.3.4 BM25 Retrieval with Grade Filtering

The physics knowledge base stored in Firestore will be filtered by the student's grade level before BM25 ranking is applied. The rank_bm25 Python library will score knowledge base documents against the extracted keyword array. The highest-scoring document will be returned as the answer if its BM25 score exceeds a defined confidence threshold. Grade filtering ensures

that a Grade 6 student asking about force receives a simplified explanation while a Grade 11 student receives a vector-based treatment of the same concept.

3.3.5 LLM Fallback

If BM25 retrieval does not produce a result above the confidence threshold, the student question and extracted keywords will be passed to the Ollama LLM as a physics explanation request with grade-level context in the prompt. The LLM response will be clearly marked as a general explanation rather than curriculum-specific content. This prevents the agent from ever returning a blank response to the student while maintaining transparency about answer provenance.

3.4 Knowledge Base Construction

The physics knowledge base is a manually curated Firestore collection constructed from Sri Lankan Grade 6-11 physics textbooks, teacher guides, and curriculum materials. Each entry covers one physics concept at one grade band, with fields for concept name, grade minimum, grade maximum, plain-language explanation, formula where applicable, and keyword tags. The prototype target is approximately 60 entries covering 20 core concepts across three grade bands. This artefact is an original contribution and will be described and evaluated as such in the research report.

3.5 Evaluation Strategy

Evaluation will be conducted across three dimensions. Pipeline functional testing will verify correct operation of each stage including RSS ingestion, classification, mapping, generation, storage, retrieval, and fallback. Output quality assessment will evaluate generated scenario questions for curriculum alignment, grade appropriateness, factual correctness, and structural completeness using a rubric applied by the researcher and supervisor. Where student participants can be recruited for a pilot session, usability evaluation will assess Q&A agent response relevance and student satisfaction using a short post-session questionnaire. The following benchmarks are intended targets subject to experimental outcomes:

Evaluation Dimension	Planned Metric	Intended Target
Physics relevance classification accuracy	Precision and Recall on held-out article set	Precision >= 0.80, Recall >= 0.75
Scenario question curriculum alignment	Rubric score by supervisor review	>= 80% of questions rated aligned
Grade appropriateness of generated questions	Rubric score by grade band	>= 75% rated appropriate for target grade
BM25 retrieval relevance	Manual relevance assessment on 30 student questions	>= 70% of top results rated relevant
Q&A agent response time	Wall-clock time per query	Response within 8 seconds on development hardware

Sinhala translation accuracy	Human evaluation of 20 Sinhala queries	>= 75% rated accurately translated

Table 3: Planned Evaluation Metrics and Intended Benchmarks

3.6 Ethical Considerations

All news content will be collected from publicly available RSS feeds. No personally identifiable data will be collected from students during evaluation. Student participation in pilot usability testing will be voluntary, and no performance data will be stored beyond anonymized session records for research purposes. The physics knowledge base will be derived entirely from publicly available Sri Lankan national curriculum materials.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

https://drive.google.com/drive/folders/1AoqKUZJpmGlim0gKvOriDJBq1TOY0ziy

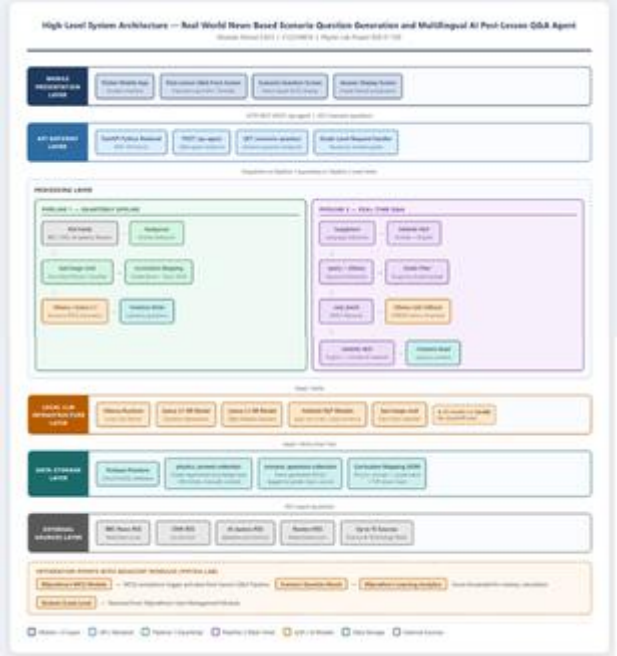


Figure 1:System_Architecture

4.1 Project Structure and Component Scope

The overall PhysiV-Lab project consists of four team-level components: (1) Grade-Adaptive RAG-Based AI Chatbot and Structured Learning Navigation (Sachintha Nirmal S.M), (2) Real-World Scenario Question Generation and Post-Lesson Q&A Agent (this component, Nirmal S.M.S), (3) Adaptive User Management and Learning Performance Evaluation (Wijerathna C.D.W), and (4) Interactive Physics Experimentation and Learning (Vidurangi H.I.D.G). This proposal describes component 2 only.

Within this component, two internal processing pipelines are implemented. The Scenario Question Pipeline operates offline on a quarterly basis and produces a stored question bank in Firestore. The Q&A Agent Pipeline operates in real time during the student lesson session. Both pipelines share the physics knowledge base and Ollama LLM infrastructure.

4.2 Component Position in Lesson Workflow

Within the lesson workflow, this component activates after Wijerathna's MCQ assessment module completes. The Q&A agent session runs first, allowing the student to resolve conceptual confusion from the MCQ phase. Once the student proceeds, the scenario question is served as the final lesson activity. Lesson completion is recorded by Wijerathna's module upon scenario question submission. The component receives the student's grade level and current lesson topic from the system's shared student profile and passes scenario question results back to the performance evaluation module.

4.3 Data Flow

In the quarterly pipeline, RSS feeds are ingested and parsed, article text is classified by the zero-shot model, curriculum-mapped articles are passed to the LLM generator, and generated questions are written to the scenario_questions Firestore collection. In the real-time Q&A pipeline, student question text is received by the FastAPI backend, processed through language detection, translation if required, keyword extraction, and BM25 retrieval or LLM fallback, and the answer text is returned to the Flutter application for display. The Flutter client communicates with the Python FastAPI backend via two endpoints: POST /qa-agent for Q&A queries and GET /scenario-question for lesson-end question retrieval.

4.4 Scalability and Extensibility

The news collection pipeline is extensible to additional RSS sources through configuration only, with no changes to core processing logic. The curriculum mapping artefact can be extended with new topic entries to support additional grade levels or subject areas. The knowledge base can be expanded incrementally. The architecture is modular, with the scenario question pipeline and Q&A agent pipeline operating independently so that either can be modified without affecting the other.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role	Primary Interest
Grade 6-11 Students (Rural)	Primary	Accessible contextual questions and on-demand concept explanations after lesson MCQs
Physics Teachers	Secondary	Curriculum-aligned scenario questions that extend classroom learning with real-world context
System Integration Team	Primary	Correct data formatting and smooth API communication between lesson components
Research Supervisors	Academic	Evidence of novelty, valid methodology, and measurable evaluation outcomes
Educational Authorities	Secondary	Curriculum alignment and potential adoption for national digital learning initiatives

Table 4:Stakeholder_Indentification

5.2 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall collect news articles from up to fifteen configured RSS feed sources on a quarterly schedule.	Must Have
FR-02	The system shall classify each collected article for physics relevance using zero-shot NLP and discard non-relevant articles.	Must Have
FR-03	The system shall map physics-relevant articles to grade bands and lesson topics using the curriculum mapping artefact.	Must Have
FR-04	The system shall generate a structured four-option MCQ scenario question for each mapped article using a locally deployed LLM.	Must Have
FR-05	The system shall store generated scenario questions in Firestore tagged with grade range, lesson topic, source, and date.	Must Have
FR-06	The system shall serve a grade-matched and topic-matched scenario question to the student at the end of each lesson.	Must Have
FR-07	The system shall serve a fallback scenario question when no generated question matches the current lesson and grade.	Must Have
FR-08	The Q&A agent shall activate only after the student has completed the lesson MCQ assessment.	Must Have
FR-09	The Q&A agent shall accept free-text student questions in English or Sinhala.	Must Have
FR-10	The system shall detect question language and translate Sinhala input to English before processing.	Must Have

FR-11	The system shall extract physics keywords from the student question using NLP and LLM-assisted extraction.	Must Have
FR-12	The system shall retrieve the most relevant knowledge base entry for the student's grade using BM25 ranking.	Must Have
FR-13	The system shall generate an LLM-based answer when BM25 retrieval does not exceed the confidence threshold.	Must Have
FR-14	The system shall translate the answer back to Sinhala when the student question was submitted in Sinhala.	Must Have
FR-15	The student shall be able to submit unlimited questions during the Q&A session.	Must Have
FR-16	The Q&A session shall be mandatory before the scenario question is presented to the student.	Should Have

Table 5:Functional_Requirements

5.3 Non-Functional Requirements

ID	Requirement	Intended Target
NFR-01	The Q&A agent shall return an answer within an acceptable response time on development hardware.	Response within 8 seconds per query
NFR-02	The physics relevance classifier shall operate without internet access during classification.	Fully offline local model execution
NFR-03	The LLM inference shall run locally without cloud API dependency.	Ollama local model only
NFR-04	The system shall handle missing or malformed RSS feeds without terminating the collection pipeline.	Zero unhandled pipeline terminations
NFR-05	The knowledge base shall support grade-filtered queries without full collection scans.	Grade-indexed Firestore queries
NFR-06	New RSS sources shall be addable through configuration only without modifying core pipeline logic.	Configuration-only source addition
NFR-07	The Q&A agent shall never return a blank response to the student.	LLM fallback active for all queries

Table 6:Non_Functional_Requirments

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market is Grade 6-11 students in rural and semi-urban government schools in Sri Lanka who have access to Android smartphones but limited laboratory and instructional support. Secondary markets include private tutoring centers, educational technology providers seeking curriculum-aligned content generation capabilities, and government education authorities interested in scalable digital learning tools for national examination preparation.

6.2 Value Proposition

The proposed component offers two distinct value propositions. For students, it provides the only mobile physics learning tool that connects real global news events to their physics curriculum through automated grade-adapted scenario questions, while also offering on-demand multilingual concept explanation without requiring a separate application or internet search. For educational institutions, it provides a zero-cost, locally deployable pipeline that continuously refreshes physics assessment content from real-world events, reducing reliance on static question banks that quickly become outdated.

6.3 Revenue Model

Should the overall PhysiV-Lab platform be commercialized, this component supports a freemium model in which the core Q&A agent and scenario question delivery are available at no cost, while premium features such as expanded knowledge bases, teacher-facing analytics on frequently asked student questions, and customized news source configurations are offered through institutional subscription. Educational authorities may license the platform for national deployment under a tiered government pricing arrangement.

6.4 Competitive Advantage

No existing mobile physics platform for Sri Lankan school students offers automated real-world news-driven scenario question generation. The multilingual Q&A agent with Sinhala support addresses a language access barrier unmet by internationally developed AI tutoring tools. The fully local processing architecture requires no cloud API subscription, making it suitable for low-resource deployment contexts where recurring API costs are prohibitive.

6.5 Intellectual Property Considerations

The curriculum mapping artefact and grade-segmented physics knowledge base constructed for this research represent original intellectual contributions specific to the Sri Lankan secondary physics curriculum. The news-to-curriculum mapping pipeline architecture may represent a novel software design if validated. Appropriate software copyright and licensing will be established in consultation with the university's commercialization guidelines following academic publication.

7. BUDGET AND JUSTIFICATION

Category	Item	Qty	Unit (LKR)	Total (LKR)
Software	All planned software (open-source: Python, spaCy, HuggingFace, Ollama, feedparser, rank_bm25, Flutter)	—	0	40000
Cloud Services	Firebase Firestore (free tier sufficient for prototype)	—	0	35000
Compute	LLM inference on researcher's personal workstation	—	0	5000
Hardware	Personal laptop for development and offline model inference	—	0	10000
Contingency	Approximately 10 percent buffer for unforeseen expenses	—	—	0
TOTAL				LKR 90000

Table 7:Budget_Summary

Justification

- All software components including Python, spaCy, HuggingFace transformers, Ollama, feedparser, rank_bm25, langdetect, Helsinki-NLP models, FastAPI, and Flutter are have to purchase licenses at RS:30000/=
- LLM inference for both quarterly question generation and real-time Q&A is performed using Ollama on the researcher's existing workstation, eliminating cloud compute expenditure entirely.
- Firebase Firestore's free tier provides sufficient storage capacity for the prototype knowledge base and scenario question collection without incurring service charges.
- The knowledge base and curriculum mapping artefacts are constructed manually from publicly available Sri Lankan national curriculum materials, incurring no data acquisition cost.

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Planned Activity	Planned Deliverable	Est. Duration
1.0	Requirements and Design		
1.1	Define RSS source list, feed URLs, and collection schema	Source configuration document	3 days
1.2	Construct curriculum mapping artefact for Grades 6-11 physics topics	Curriculum mapping JSON file	1 week
1.3	Define knowledge base schema and target entry list	Knowledge base specification	3 days
1.4	Design Q&A agent API interface contracts with adjacent modules	Interface specification	2 days
2.0	Artefact Construction		
2.1	Populate physics knowledge base from curriculum materials (20 concepts, 3 grade bands)	60-entry knowledge base in Firestore	1.5 weeks
2.2	Construct and validate curriculum mapping table entries	Validated mapping artefact	4 days
3.0	Scenario Question Pipeline Development		
3.1	Implement RSS collection and article storage module	Working collection script	4 days
3.2	Implement and evaluate zero-shot physics relevance classifier	Classifier with evaluation results	1 week
3.3	Implement curriculum mapping and grade filtering module	Working mapping module	4 days
3.4	Implement and evaluate LLM scenario question generation with prompt engineering	Working generation pipeline	1 week
3.5	Integrate pipeline stages and run end-to-end on collected articles	Integrated scenario pipeline	3 days
4.0	Q&A Agent Development		
4.1	Implement language detection and Sinhala translation module	Working translation module	4 days
4.2	Implement keyword extraction using spaCy and Ollama	Working extraction module	4 days
4.3	Implement BM25 retrieval with grade filtering	Working retrieval module	4 days
4.4	Implement LLM fallback module and confidence threshold logic	Working fallback module	3 days
4.5	Implement FastAPI endpoints and integrate all Q&A agent stages	Integrated Q&A agent API	4 days
5.0	Flutter UI Development		
5.1	Implement post-lesson Q&A form screen in Flutter	Working Q&A UI screen	4 days
5.2	Implement scenario question delivery screen in Flutter	Working scenario question screen	3 days
6.0	Testing and Evaluation		

6.1	Pipeline functional testing across all stages	Functional test report	1 week
6.2	Classifier and question quality evaluation using rubric	Quality evaluation report	1 week
6.3	Usability pilot with student participants where feasible	Usability evaluation report	1 week
7.0	Documentation and Reporting		
7.1	Write research report chapter for this component	Draft report chapter	2 weeks
7.2	Prepare presentation materials	Presentation slides	3 days

Table 8:Work_Breakdown_Structure

9. GANTT CHART

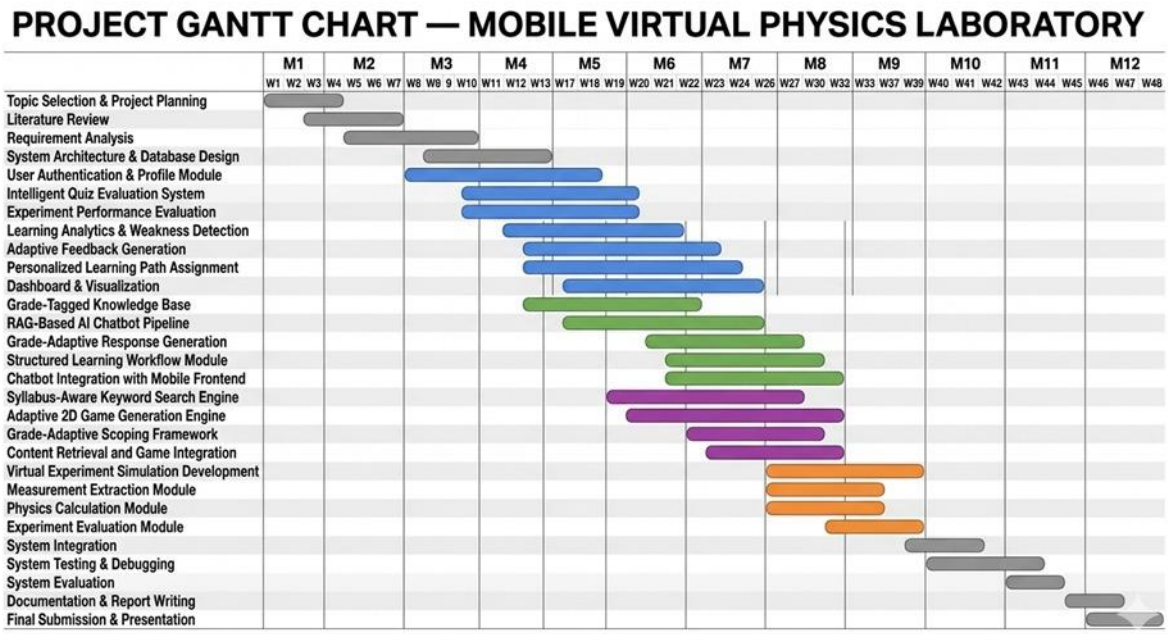


Figure 2:Gantt_chart

10.REFERENCES LIST

[1] M. A. Rahman, M. A. Hossain, and M. S. Islam, "Virtual Laboratories in STEM Education," Journal of Advanced Learning and Teaching, vol. 4, no. 2, pp. 1-15, 2021. Available: https://jalt.com.pk/index.php/jalt/article/view/1226

[2] R. Vijayatheepan, "Virtual Labs in Science Teaching: Teachers' Perspectives and Challenges in Sri Lankan Schools," Journal of Innovation in Physics Education, Open University of Sri Lanka Repository, 2025.

[3] R. Elmoazen, M. H. Al-Mamun, and A. Al-Hassan, "Learning Analytics in Virtual Laboratories: A Systematic Literature Review of Empirical Research," Smart Learning Environments, vol. 10, no. 18, 2023. doi: 10.1186/s40561-023-00244-y

[4] M. Kalogiannakis, S. Papadakis, and A. I. Zourmpakis, "Gamification in Science Education: A Systematic Review of the Literature," Education Sciences, vol. 11, no. 1, pp. 1-36, 2021. doi: 10.3390/educsci11010022

[5] A. B. Gil, A. de la Prieta, S. Rodriguez, and J. M. Corchado, "Smart System for the Retrieval of Digital Educational Content Based on Machine Learning," Applied Sciences, vol. 9, no. 20, 2019. doi: 10.3390/app9204400

[6] W. Huang, K. F. Hew, and B. Lo, "Investigating the Effects of AI Chatbots on Students' Learning Outcomes and Motivation," Educational Technology & Society, vol. 22, no. 2, pp. 1-13, 2019.

[7] O. Zawacki-Richter, V. I. Marin, M. Bond, and F. Gouverneur, "Systematic Review of Research on Artificial Intelligence Applications in Higher Education," International Journal of Educational Technology in Higher Education, 2019. doi: 10.1186/s41239-019-0171-0

[8] X. Zhai, X. Chu, C. S. Chai, M. S. Y. Jong, A. Istenic, M. Spector, J. Liu, J. Yuan, and Y. Li, "A Review of Artificial Intelligence (AI) in Education from 2010 to 2020," Complexity, vol. 2021, pp. 1-18, 2021. doi: 10.1155/2021/8812542

[9] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, M. Lewis, W.-t. Yih, T. Rocktaschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," Advances in Neural Information Processing Systems, vol. 33, pp. 9459-9474, 2020.

[10] S. E. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," Foundations and Trends in Information Retrieval, vol. 3, no. 4, pp. 333-389, 2009. doi: 10.1561/1500000019

[11] M. Hammami, Y. Chahir, and L. Chen, "WebGuard: A Web Content Mining-Based System for Automatic and Intelligent Web Filtering," IEEE Transactions on Knowledge and Data Engineering, vol. 18, no. 2, pp. 272-284, Feb. 2006. doi: 10.1109/TKDE.2006.34

[12] I. Yin, Z. Zheng, Z. Zu, X. Zeng, and B. Hu, "A Survey of Zero-Shot Text Classification: Towards a Taxonomy and Unified View," arXiv preprint arXiv:2306.00649, 2023.

[13] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design Science in Information Systems Research," MIS Quarterly, vol. 28, no. 1, pp. 75-105, Mar. 2004. doi: 10.2307/25148625

11. APPENDICES

Appendix A: Preliminary Feasibility Study

To assess technical viability before committing to full implementation, a preliminary investigation was conducted into the availability and quality of physics-relevant content in international news RSS feeds. A sample collection of 50 articles was manually gathered from BBC and Al Jazeera RSS feeds and reviewed against the Sri Lankan Grade 6-11 physics curriculum. Preliminary observations indicate that approximately 18 to 25 percent of collected science and technology articles contain content mappable to one or more curriculum topics, with motion and forces, energy, and electricity being the most frequently represented categories. This suggests that a quarterly collection of 200 to 300 articles would yield approximately 40 to 75 physics-relevant articles per cycle, providing a sufficient volume of scenario questions. These are preliminary observations based on manual review only and should not be interpreted as validated classifier performance metrics.

Appendix B: Physics Knowledge Base Sample Entries

The following illustrates the planned structure and content level for knowledge base entries at different grade bands for the concept of velocity:

Concept	Grade Band	Explanation
Velocity	6-7	Velocity is how fast something moves in a specific direction. If a car travels 60 km in one hour heading north, its velocity is 60 km/h north. Velocity is different from speed because direction matters.
Velocity	8-9	Velocity is a vector quantity defined as the rate of change of displacement. It is calculated as v = s/t, where s is displacement in metres and t is time in seconds. Unlike speed, velocity specifies the direction of motion.
Velocity	10-11	Velocity is a vector quantity representing the time derivative of displacement: v = ds/dt. Average velocity is calculated over a time interval, while instantaneous velocity is the limit of displacement over an infinitesimally small time interval. Velocity appears in Newton's Second Law through momentum p = mv.

Table 9:Explanation_01

Appendix C: Risk Analysis

Risk	Likelihood	Impact	Mitigation
RSS feed URL changes or becomes unavailable	Medium	Low	Maintain a list of alternative feeds per source; configuration-only update required
Zero-shot classifier produces low precision on physics articles	Medium	Medium	Tune confidence threshold; supplement with keyword pre-filter based on physics vocabulary list
LLM generates structurally invalid JSON question output	Medium	Medium	Implement output validation with retry logic; maintain fallback hand-authored question set

Ollama model too slow for real-time Q&A on low-spec hardware	Medium	High	Use Phi-3 Mini or Mistral 7B as lighter alternative; cache frequent query responses
Sinhala translation quality insufficient for clear physics explanations	Medium	Medium	Human review of translated outputs during evaluation; fallback to English display with Sinhala label if translation confidence is low
Insufficient pilot participants for usability evaluation	Low	Medium	Conduct evaluation with researcher and supervisor as proxy users; supplement with rubric-based output quality assessment

Table 10:Explanation_02

Appendix D: Curriculum Mapping Table Sample

The following illustrates a sample of the curriculum mapping artefact entries

Physics Concept Tag	Grade Band	Lesson Topic	Example News Event Type
Motion and Forces	6-7	Speed, Distance, Time	Vehicle accidents, sports records, train journeys
Motion and Forces	8-9	Newton's Laws of Motion	Rocket launches, car crashes, sports biomechanics
Motion and Forces	10-11	Momentum and Projectile Motion	Missile trajectories, satellite launches, ballistics
Electricity	8-9	Current, Voltage, Circuits	Power outages, renewable energy projects, electric vehicles
Energy and Waves	6-7	Forms of Energy	Solar panels, wind turbines, earthquake energy
Pressure and Fluids	9-11	Fluid Pressure, Buoyancy	Submarine operations, dam failures, weather pressure systems

Table 11: Explanation_03